

Network Layer

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- ❑ Network Layer Design issues, Functions of Network layer.
 - ❑ Switching Techniques: Circuit switching, Message Switching, Packet Switching.
 - ❑ IP Protocol: Classes of IP, IPv4, IPv6, NAT, Sub-netting, CIDR.
 - ❑ Network layer Protocols: ARP, RARP, ICMP, IGMP.
 - ❑ Network Routing Concepts: Static Routing, Dynamic Routing, Distance Vector Routing, Link State Routing, Path Vector.
 - ❑ Routing Protocols: RIP, OSPF, BGP, IGRP. Quality of Service

Purpose of Network Layer

- ❑ The main purpose of the network layer is to transmit packets from one device to another.
- ❑ It contains the complete details of the router, which decide the optimal path to send the packets.
- ❑ The network layer sends packets in such a way that there is less congestion on the network.
- ❑ It uses IP, messaging, and routing protocols for communication. The IP address is used to transmit the packets, and the routing protocol is used to find the best path for the transmitting packets. Messaging protocol such as ICMP is used as reply and request protocol.
- ❑ The network layer communicates with the transport layer and the data link layer. It receives segments from the transport layer and converts them into packets.
- ❑ When the network layer creates the packet, it includes the sender and receiver device IP address so that the router can find the receiver and send it correctly.

Functions of Network layer

- ❑ To provide end-to-end communication, the network layer performs basic operations such as encapsulation, forwarding, and addressing, etc.
- ❑ The network layer has the following responsibilities:
 - Encapsulation and de-encapsulation: Encapsulates segments on the sender side and de-encapsulates the frames on the receiver side.
 - IP Addressing: The IP addresses of the sender and receiver are stored in the packet header.
 - Connectionless services: No dedicated path is established between devices in advance for communication.
 - Routing Decision: Finds the best route to send packets.

Encapsulation and De-encapsulation

❑ Encapsulation:

- The network layer communicates with the transport layer and receives a segment.
- When the network layer receives the segment, it adds the IP header to the segment that makes up the packet.
- This process is known as encapsulation. Packets in the network layer are also known as Network Layer Protocol Data Units (PDUs).

❑ De-encapsulation:

- When a packet sent by the sender is received to the receiver, it first goes to the physical layer as a stream of bits and then to the data link layer as a frame.
- As the data link layer sends a frame to the network layer, the network layer removes the frame header and trailer that makes up the packet. This process is known as de-encapsulation.

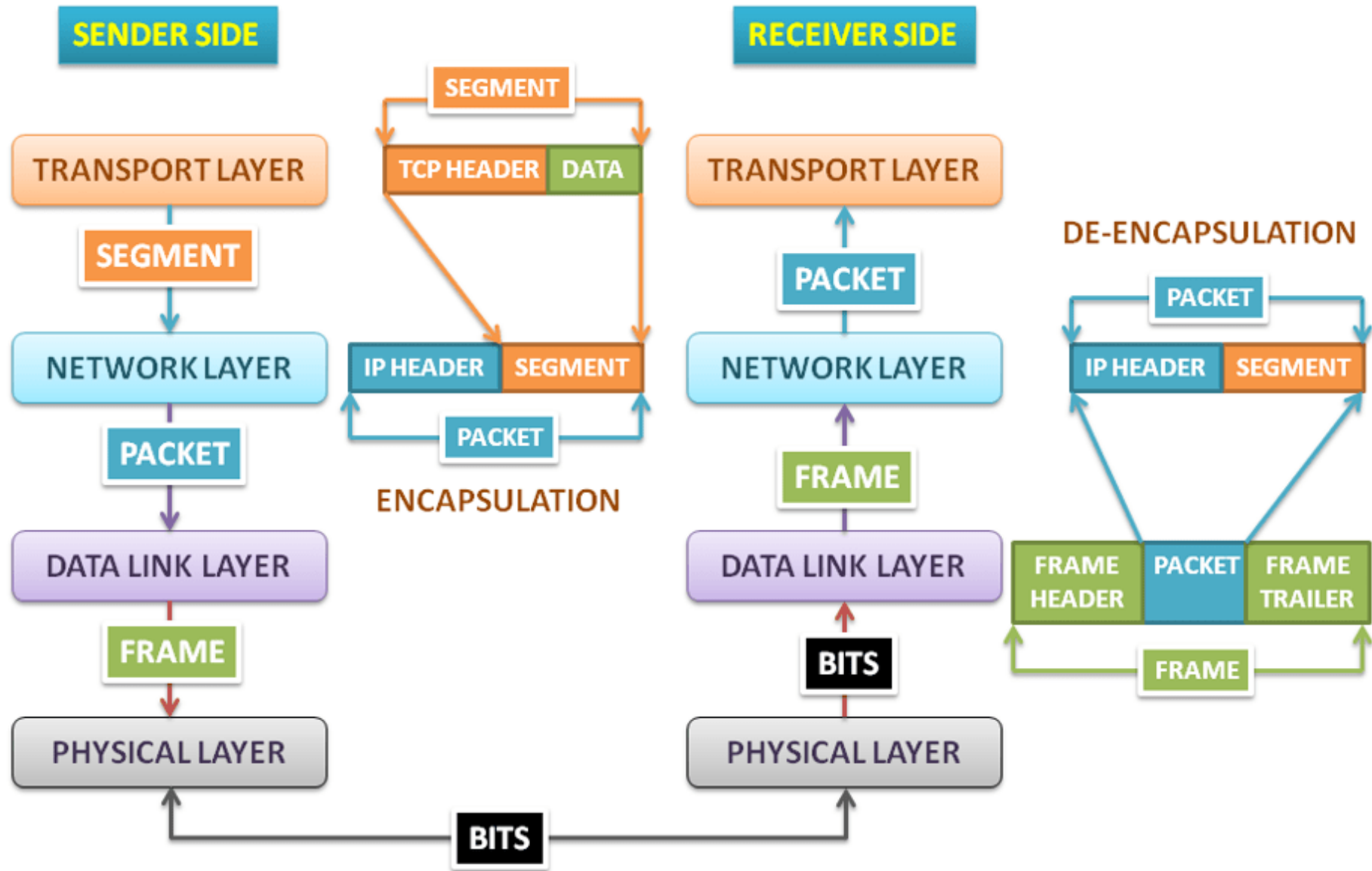


Figure 2: Encapsulation and De-Encapsulation of Packet



IP Addressing

- ❑ The network layer only understands the IP addressing scheme. It uses IP protocol for communication, and IP does not guarantee that all packets will be delivered correctly. This is known as a best-effort mechanism.
- ❑ Internet Protocol (IP) is used at the network layer and provides the operations needed to transmit a packet from sender to receiver through multiple routers.
- ❑ Also, IP does not track packets or manage the flow of packets. Flow control is done using layer-4 protocols like TCP.
- ❑ IP does not depend on the type of media. It does not matter whether the media is wireless or wired. It just takes the data and sends it to the receiver.
- ❑ IP addresses have versions IPv4 (version 4) and IPv6 (version 6). IPv4 and IPv6 are 32-bit and 128-bit in size, respectively.
- ❑ The IP address information is stored in the IP header field of the packet.
- ❑ Typically, IP addresses are used to make decisions on the network to find the destination for transmitted packets using a best-effort mechanism.

Connectionless Services

- ❑ In the network layer, IP uses a connectionless service which means there is no need to establish a dedicated path between the sender and the receiver before sending the data.
- ❑ For a connectionless service, it is not necessary to establish an end-to-end connection for the exchange of IP information between the sender and the receiver.
- ❑ Connectionless services use the same concept as sending a letter to the recipient. For example, you write a letter to your friend without informing him in advance and put it in the mailbox of the post office. Therefore, your friend will not know until your letter is received by him.

IP Packet Delivery

- ❑ IP does not have any other information in the header which is used for connection establishment. This property reduces the overhead of IP on a network.
- ❑ When the IP packet travels over the communication channel, it does not know whether the receiver is ready to receive the packet.
- ❑ Furthermore, if the receiver receives an IP packet, the sender cannot decide whether the IP packet has been successfully received by the recipient.
- ❑ This characteristic of IP is known as a best-effort mechanism.
- ❑ It is necessary to use some protocols on the network to track the information of IP packets. That's why the network layer communicates with the transport layer and uses the TCP protocol to track the information of packets sent to the receiver.

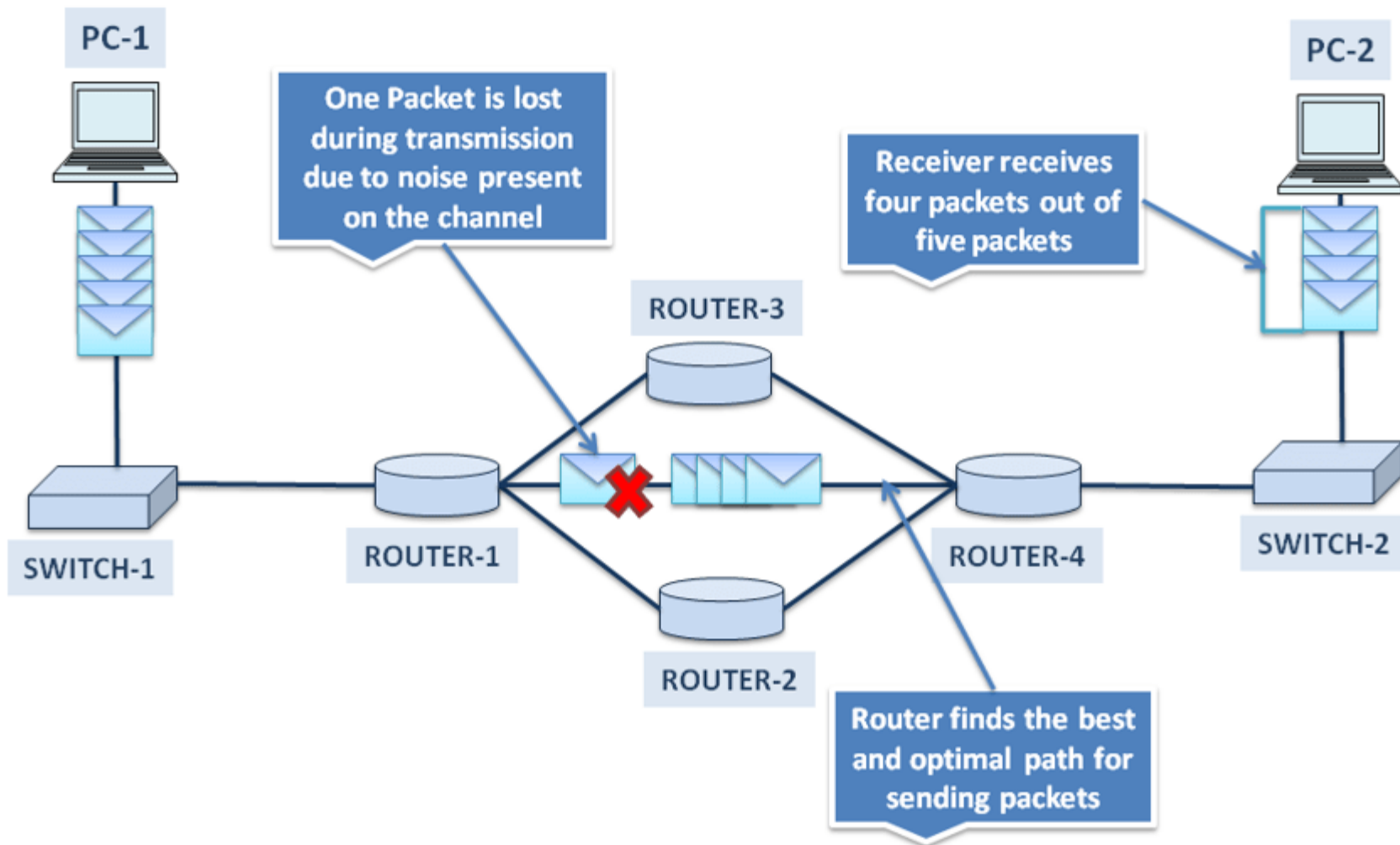


Figure 4: Delivery of Packets using Best-Effort Mechanism



Routing Decision

- ❑ The decision to send packets is made in several ways. For example, a device may send packets to itself or another device that is on the same LAN or a device that is on another LAN.
- ❑ If the two device IP addresses are in the same LAN, the sender will send the packet directly to the receiver through an intermediary device.
- ❑ But in a real network, the packet has to be sent to the receiver, located far away from the sender. So the router works in the network layer and uses the services of the network layer to build the routing table and decide the total hops from sender to receiver.
- ❑ A router has a gateway through which packets are transmitted from one network to another. In simple words, the router routes the traffic to another network.

Network Layer Design issues

❑ Store-and-Forward Packet Switching

- The network layer follows store-and-forward mechanism where routers receive the complete packet, store it temporarily, and then forward it to the next hop. This ensures error checking and proper routing before transmission.

❑ Services Provided to the Transport Layer

- The network layer must provide reliable and efficient logical communication between source and destination. It defines whether the service is connection-oriented (virtual circuit) or connectionless (datagram).

❑ Implementation of Connectionless Service (Datagram Model)

- In the datagram approach, each packet is treated independently and may follow different routes to reach the destination. Example: Internet Protocol (IP).

❑ Implementation of Connection-Oriented Service (Virtual Circuit Model)

- A logical path is established before data transmission begins, and all packets follow the same path. Resources can be reserved for better Quality of Service (QoS).

❑ Routing Algorithms

- Determines the best path for packet delivery from source to destination. Routing can be static or dynamic depending on network conditions.

❑ Congestion Control

- Mechanisms are required to prevent network overload when too many packets are present. Proper congestion control improves throughput and reduces packet loss.

❑ Quality of Service (QoS)

- The network layer must support parameters like bandwidth, delay, jitter, and reliability for real-time applications such as video streaming and VoIP.

❑ Internetworking

- Connecting multiple heterogeneous networks into a single unified network. Protocols like IP enable communication across different network technologies.

❑ Addressing

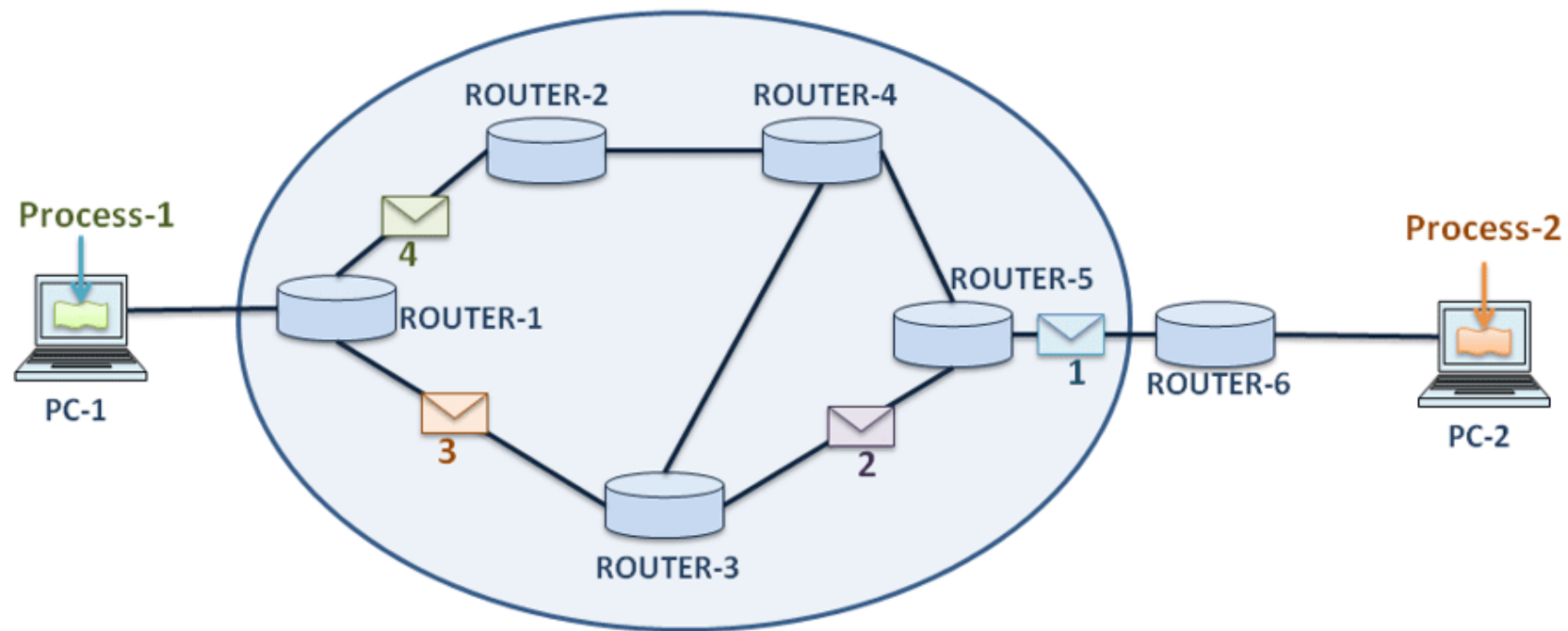
- Logical addressing (IP addressing) is used to uniquely identify devices across networks. It ensures proper packet delivery to the correct destination.

❑ Fragmentation and Reassembly

- Large packets may need to be divided into smaller fragments to pass through networks with smaller Maximum Transmission Units (MTU). These fragments are reassembled at the destination.

Connectionless Service

- ❑ The network layer provides two types of service to its users.
- ❑ In a connectionless service, packets sent by the sender are individually injected into the network that travels through different routes.
- ❑ There is no need for an advanced path setup in a connectionless service to transmit the packets.
- ❑ In connectionless service, packets are called datagrams, and the network used to transmit packets is known as datagram networks.
- ❑ Connectionless service is an unreliable service. When two devices want to communicate with each other, a predetermined path is not required.
- ❑ IP protocol at the network layer and UDP protocol at the transport layer uses connectionless service. Since they are using connectionless services, there is no guarantee of successful packet delivery.



Router-1's table initially

Destination	Line
R-1	---
R-2	R-2
R-3	R-3
R-4	R-2
R-5	R-3
R-6	R-3

Router-1's table later

Destination	Line
R-1	---
R-2	R-2
R-3	R-3
R-4	R-2
R-5	R-2
R-6	R-2

Router-3's table

Destination	Line
R-1	R-1
R-2	R-4
R-3	---
R-4	R-5
R-5	R-5
R-6	R-5

Router-5's table

Destination	Line
R-1	R-3
R-2	R-4
R-3	R-3
R-4	R-4
R-5	---
R-6	R-6

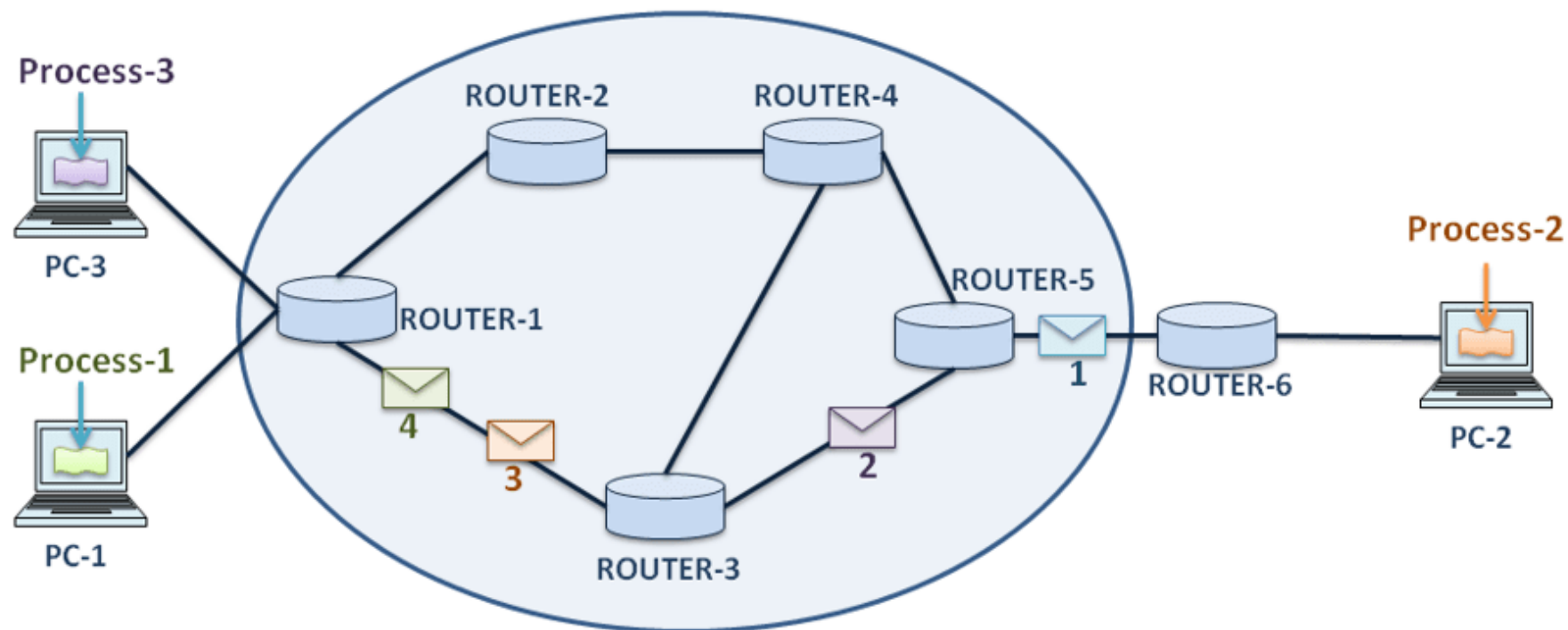
Figure 1: Implementation of Datagram Network



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- ❑ As shown in the figure, PC-1 and PC-2 wants to communicate. PC-1 needs to send the packets to PC-2. Process-1 and Process-2 are running on PC-1 and PC-2, respectively.
 - ❑ The network layer communicates with the transport layer and sends the message with instructions to deliver packets to the PC-2 process.
 - ❑ The network layer breaks the packet into four packets, packet-1, packet-2, packet-3, and packet-4.
 - ❑ All four packets are transmitted to Router-1 using the point-to-point protocol. Router-1 has an initial router table that decides where to send the packets.
 - ❑ Each router has two columns, Destination, and Line. Using the line column, the router determines the path to reach the other router. But, it uses only directly connected lines.
 - ❑ For example, Router-1 has two outgoing lines Router-2 and Router-3. So Router-1 will send each packet to Router-2 and Router-3.
 - ❑ Packet-1, Packet-2 and Packet-3 are transmitted through the path Router-1 -> Router-3 -> Router-5 -> Router-6 -> PC-2, while Packet-4 is transmitted through the path Router-1 -> Router-2 -> Router-4 -> Router-5 -> Router-6 -> PC-2.
 - ❑ But why is packet-4 traveling on a different path than the path of the first three packets? Because Router-1 understood the network and detected a traffic jam somewhere along the 1-3-5-6-PC2 way.
 - ❑ The routing algorithm is used to manage and make routing decisions based on a routing table.

Connection-Oriented Service

- ❑ In connection-oriented service, a predetermined path for communication is established before communication takes place between devices.
- ❑ The path established between sender and receiver is known as a virtual circuit, and the network used to transmit packets is known as a virtual-circuit network.
- ❑ Virtual-circuit networks are used in a connection-oriented service. Using a virtual-circuit router prevents packets from being sent on another path instead of the prescribed one.
- ❑ When a connection is established between two devices, the route information is stored in the routing table.
- ❑ The chosen route is used for sending all packets and all traffic flowing over the connection. For example, the telephone system uses a connection-oriented service.
- ❑ In a connection-oriented service, each packet has an identifier used to describe which virtual circuit it belongs to.
- ❑ Multiprotocol Label Switching (MPLS) uses a connection-oriented service. IP packets are wrapped in an MPLS header that contains a 20-bit connection identifier, which is used to identify the path to send the packet.



Virtual-Circuit
Number

Router-1's table

Dest.	No.	Dest.	No.
PC-1	1	R-3	1
PC-3	1	R-3	2

In Out

Router-3's table

Dest.	No.	Dest.	No.
R-1	1	R-5	1
R-1	2	R-5	2

Router-5's table

Dest.	No.	Dest.	No.
R-3	1	R-6	1
R-3	2	R-6	2

Figure 2: Implementation of Virtual Circuit Network



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- ❑ As shown in the figure, the connection is established before the communication between PC-1 and PC-2.
 - ❑ PC-1 is sending four packets to PC-2. Here, the identifier -1 is assigned to the packets of PC-1. Packet-1, Packet-2, Packet-3 and Packet-4 are transmitted to PC-2 with the identifier -1.
 - ❑ Now, PC-3 also wants to send a packet to PC-2. It asks for identifier-1 from the network layer to establish a virtual-circuit network. But it is already assigned to PC-2. The router-1 assigns identifier-2 to PC-3.
 - ❑ In the end, the routing table has two entries, one for PC-1 and the other for PC-3.
 - ❑ Different identifiers are used to avoid packet collisions on the network.

□ There are several issues with connectionless and connection-oriented services within the network. They are as follows:

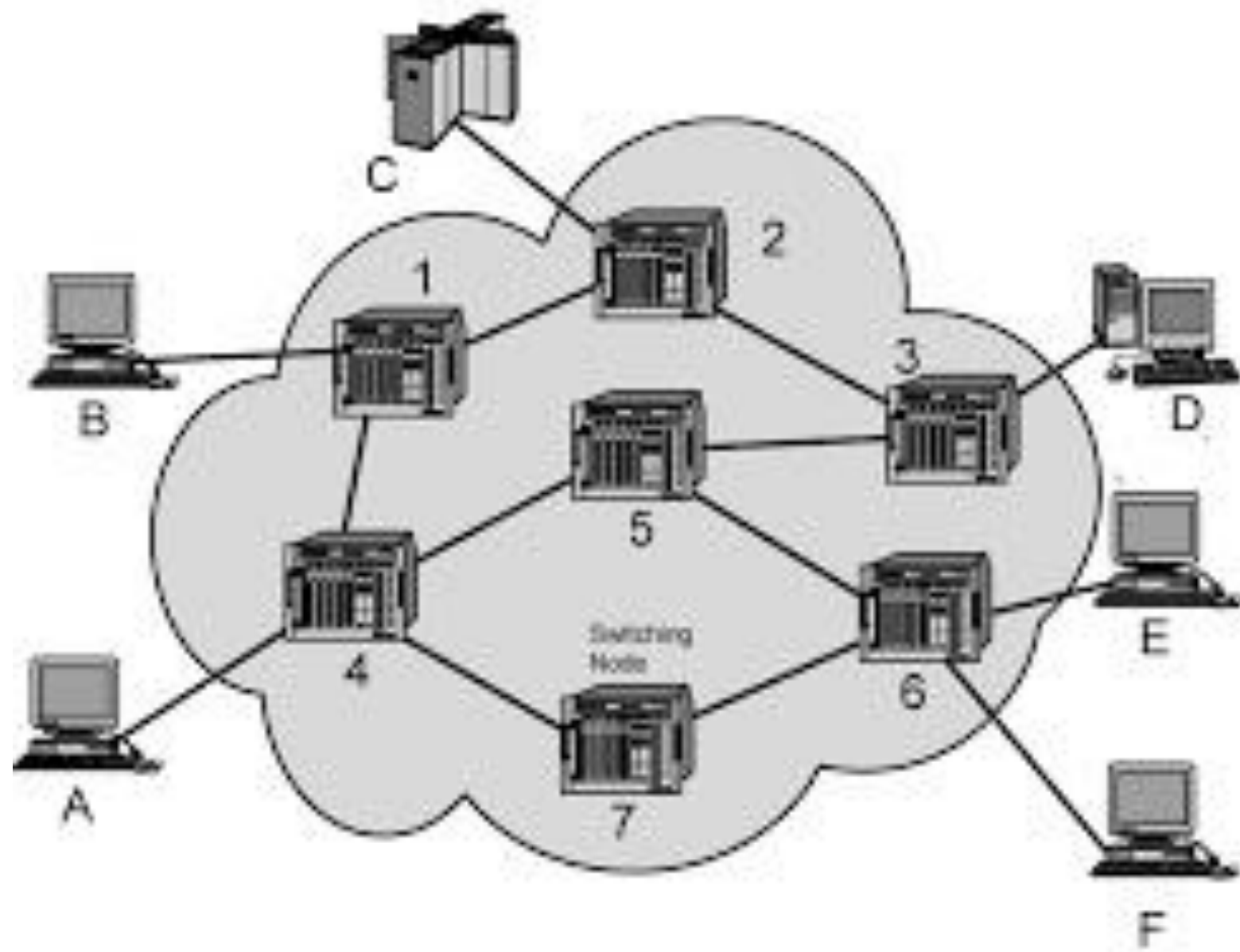
- In virtual-circuit networks, the path has to be predetermined, which takes setup time and consumes resources. But once the path is established between the devices, it is easy to decide what to do with the packets.
- The router uses the circuit number in the virtual-circuit network to find the destination to send the packet. This feature helps in figuring out what to do with the packets and where to send them.
- Communication in a datagram network does not require a predetermined path. But this strategy requires a more complex lookup process for creating routing tables and sending packets.
- Datagram networks do not use virtual-circuit networks. The router creates a destination address entry in the routing table instead of a virtual circuit. The destination address is longer than the circuit numbers used in a virtual circuit, this causes a significant amount of overhead, and hence, a waste of bandwidth.
- The router must have enough memory to store the destination address in the datagram network. Routers must have every possible destination address entry that uses more memory, whereas, in virtual-circuit networks, routers require only one entry for each virtual circuit.

Connectionless vs connection-oriented service

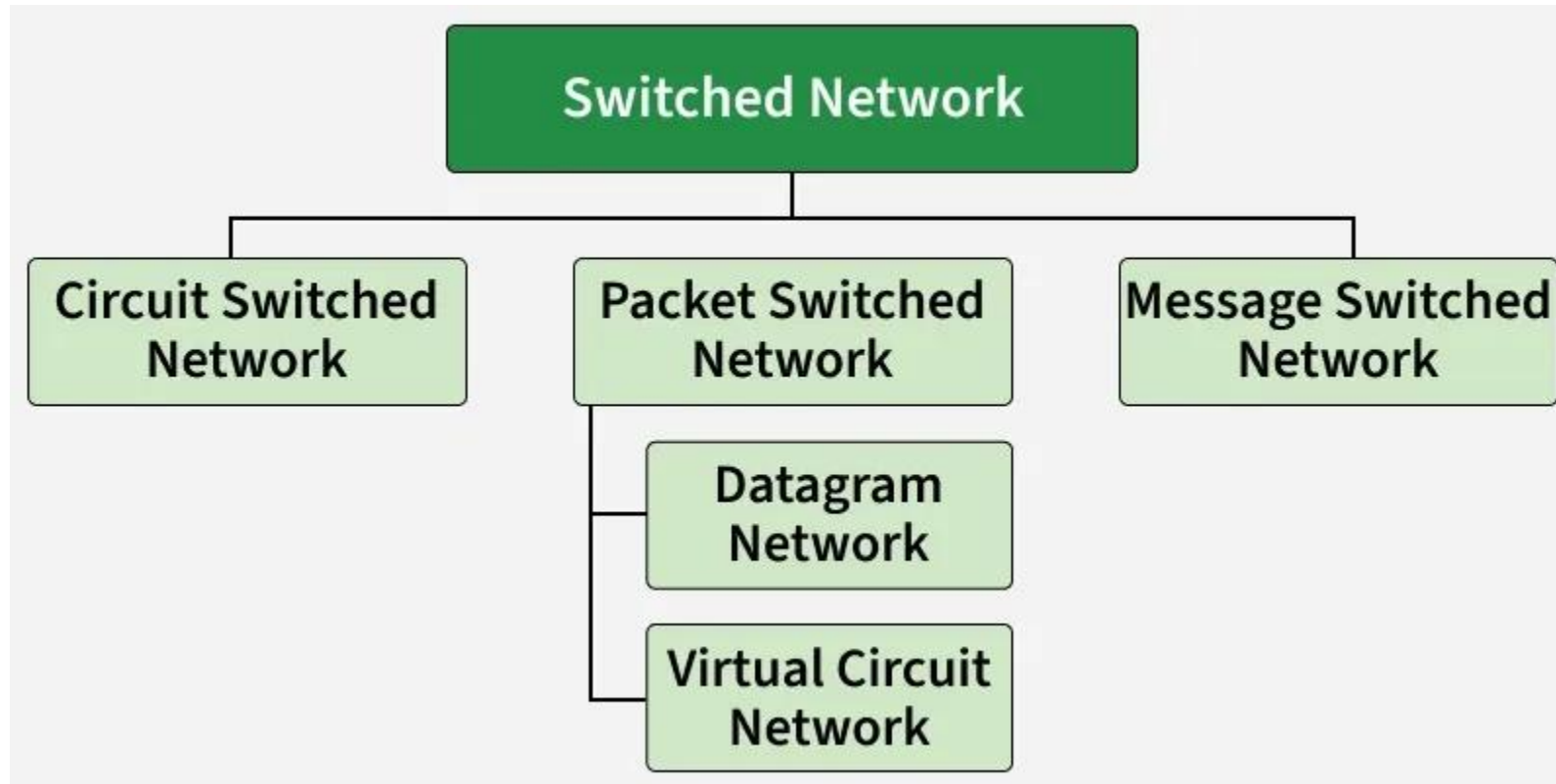
Key	Connectionless Service	Connection-Oriented Service
Path Setup	Not needed	Established in advance
Address	Each packet has a complete source and destination address.	Each packet contains a virtual circuit number.
Routing Decision	A packet is routed independently.	Packets follow the virtual circuit number.
Resources Reservation	Not reserved	Reserved before communication
Router Failure	Affects only packets lost during transmission.	Affects all VCs passing through it.
Congestion Control	Difficult	Easy
Quality of Service	Difficult	Easy
Network	Datagram Network	Virtual-Circuit Network
Protocols	IP, UDP, etc.	TCP, X.25, Frame Relay, etc.

Switching Techniques

- ❑ Switching is the process of transferring data from one device to another within the same network or across different networks using switches. It mainly occurs at the Data Link Layer (Layer 2) of the OSI model, where data is forwarded in the form of frames based on MAC addresses.
- ❑ A switch connects multiple devices within a network.
- ❑ It reduces interference by sending data only to the intended device.
- ❑ It forwards data frames to the correct port using MAC addresses.
- ❑ It delivers data directly to connected devices (computers, phones).
- ❑ For communication outside the local network, it forwards data to a router.

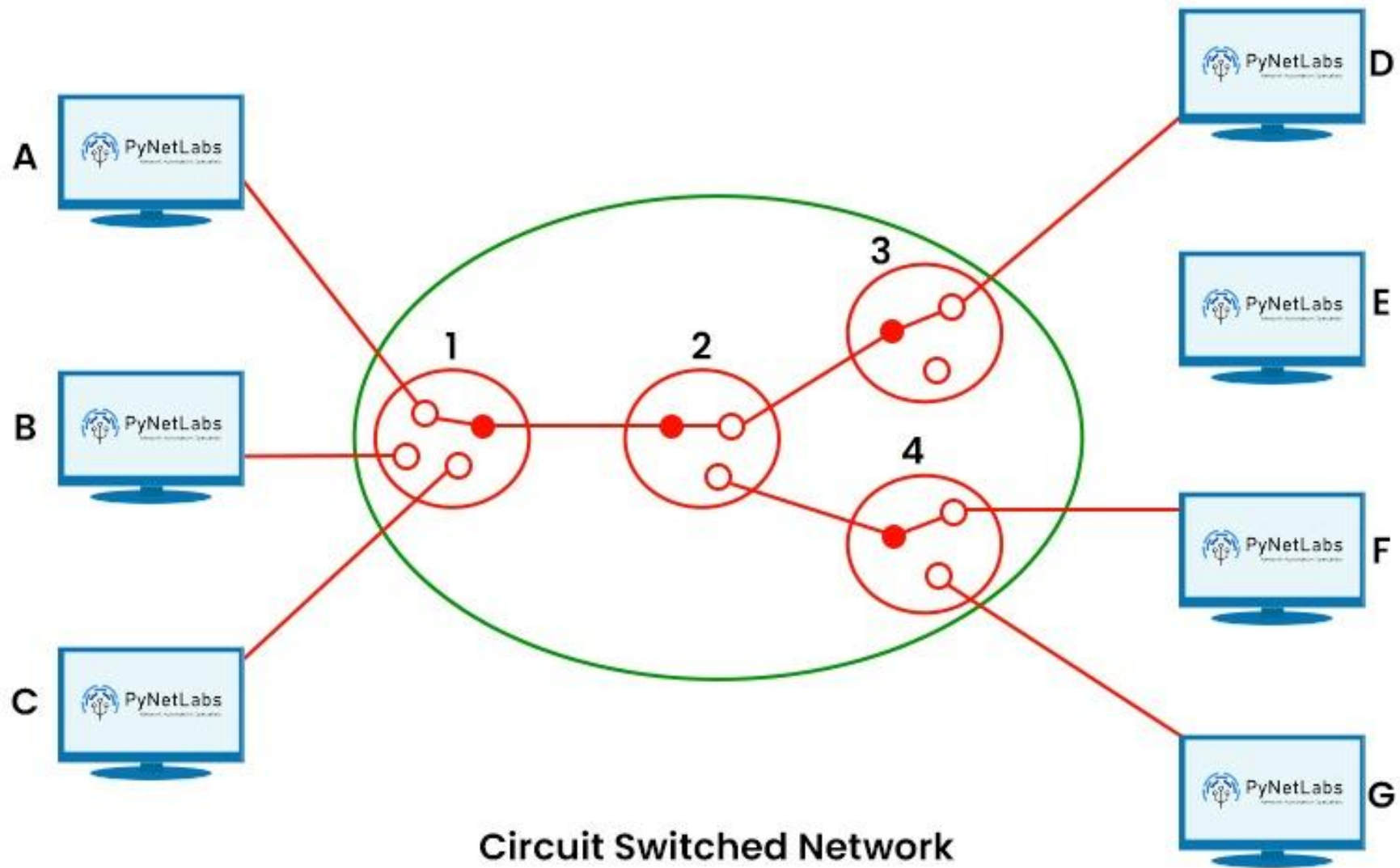


Types of Switching



Circuit Switching

- ❑ Circuit switching is a technique that directly connects the sender and the receiver in an unbroken path.
- ❑ Telephone switching equipment, for example, establishes a path that connects the caller's telephone to the receiver's telephone by making a physical connection.
- ❑ With this type of switching technique, once a connection is established, a dedicated path exists between both ends until the connection is terminated.
- ❑ Routing decisions must be made when the circuit is first established, but there are no decisions made after that time.
- ❑ Circuit switching in a network operates almost the same way as the telephone system works.
- ❑ A complete end-to-end path must exist before communication can take place.
- ❑ The computer initiating the data transfer must ask for a connection to the destination.
- ❑ Once the connection has been initiated and completed to the destination device, the destination device must acknowledge that it is ready and willing to carry on a transfer.



❑ Advantages:

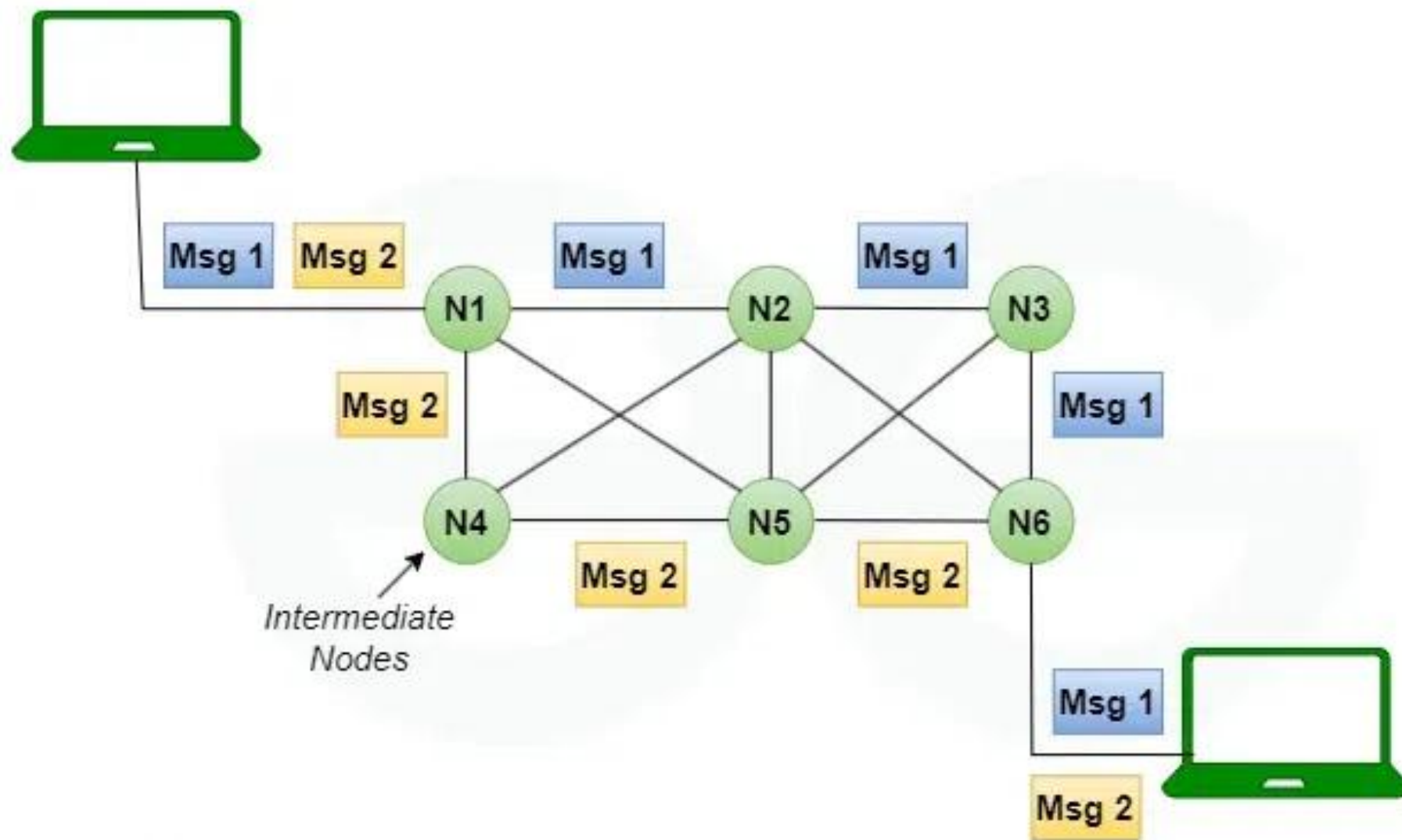
- The communication channel (once established) is dedicated.

❑ Disadvantages:

- Possible long wait to establish a connection, (10 seconds, more on long distance or international calls.) during which no data can be transmitted.
- More expensive than any other switching techniques, because a dedicated path is required for each connection.
- Inefficient use of the communication channel, because the channel is not used when the connected systems are not using it.

Message Switching

- ❑ With message switching there is no need to establish a dedicated path between two stations.
- ❑ When a station sends a message, the destination address is appended to the message.
- ❑ The message is then transmitted through the network, in its entirety, from node to node.
- ❑ Each node receives the entire message, stores it in its entirety on disk, and then transmits the message to the next node.
- ❑ This type of network is called a store-and-forward network.
- ❑ A message-switching node is typically a general-purpose computer.
- ❑ The device needs sufficient secondary-storage capacity to store the incoming messages, which could be long.
- ❑ A time delay is introduced using this type of scheme due to store- and-forward time, plus the time required to find the next node in the transmission path.



❑ Advantages

- Channel efficiency can be greater compared to circuits switched systems, because more devices are sharing the channel.
- Traffic congestion can be reduced, because messages may be temporarily stored in route.
- Message priorities can be established due to store-and-forward technique.
- Message broadcasting can be achieved with the use of broadcast address appended in the message.

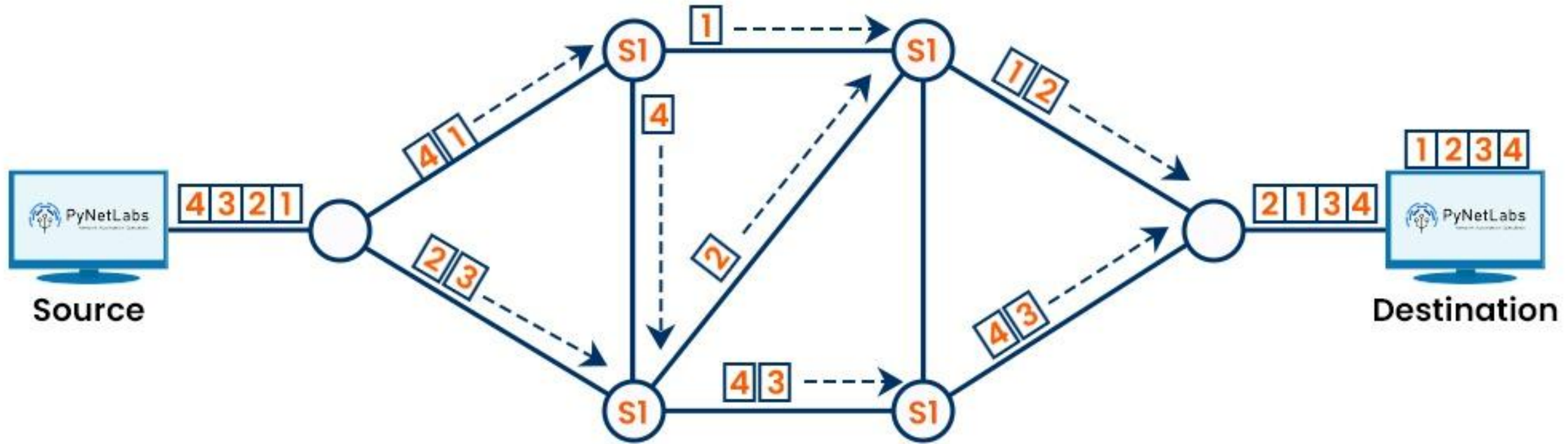
❑ Disadvantages

- Message switching is not compatible with interactive applications.
- Store-and-forward devices are expensive, because they must have large disks to hold potentially long messages.

Packet Switching

- ❑ Packet switching is a type of connectionless or connection-oriented switching technique that divides the message into units of variable length known as packets, each of which is given a source and destination address.
- ❑ Without buffering or sequencing, the switches independently forward the packets.
- ❑ Depending on the network configuration and the availability of pathways, the packets may travel different paths to reach their destination.
- ❑ The packets are put back together by the destination device to form the original message.
- ❑ Packet switching is suitable for applications that require efficient and flexible communication, such as web browsing or streaming.

Packet Switching



Datagram packet switching

- ❑ Datagram packet switching is similar to message switching in that each packet is a self-contained unit with complete addressing information attached.
- ❑ This allows packets to take a variety of possible paths through the network.
- ❑ So, the packets, each with the same destination address, do not follow the same route, and they may arrive out of sequence at the exit point node (or the destination).
- ❑ Reordering is done at the destination point based on the sequence number of the packets.
- ❑ It is possible for a packet to be destroyed if one of the nodes on its way is crashed momentarily. Thus all its queued packets may be lost.

Virtual Circuit

- ❑ In the virtual circuit approach, a preplanned route is established before any data packets are sent.
- ❑ A logical connection is established when
 - a sender send a "call request packet" to the receiver and
 - the receiver send back an acknowledge packet "call accepted packet" to the sender if the receiver agrees on conversational parameters.
- ❑ The conversational parameters can be maximum packet sizes, path to be taken, and other variables necessary to establish and maintain the conversation.
- ❑ Virtual circuits imply acknowledgements, flow control, and error control, so virtual circuits are reliable.
- ❑ That is, they have the capability to inform upper-protocol layers if a transmission problem occurs.

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- ❑ In virtual circuit, the route between stations does not mean that this is a dedicated path, as in circuit switching.
 - ❑ A packet is still buffered at each node and queued for output over a line.
 - ❑ The difference between virtual circuit and datagram approaches:
 - With virtual circuit, the node does not need to make a routing decision for each packet.
 - It is made only once for all packets using that virtual circuit.
 - ❑ VC guarantees that
 - the packets arrive in the order they sent
 - there are no duplicates or omissions.
 - there are no errors (with high probability) regardless of how they are implemented internally.

Advantages

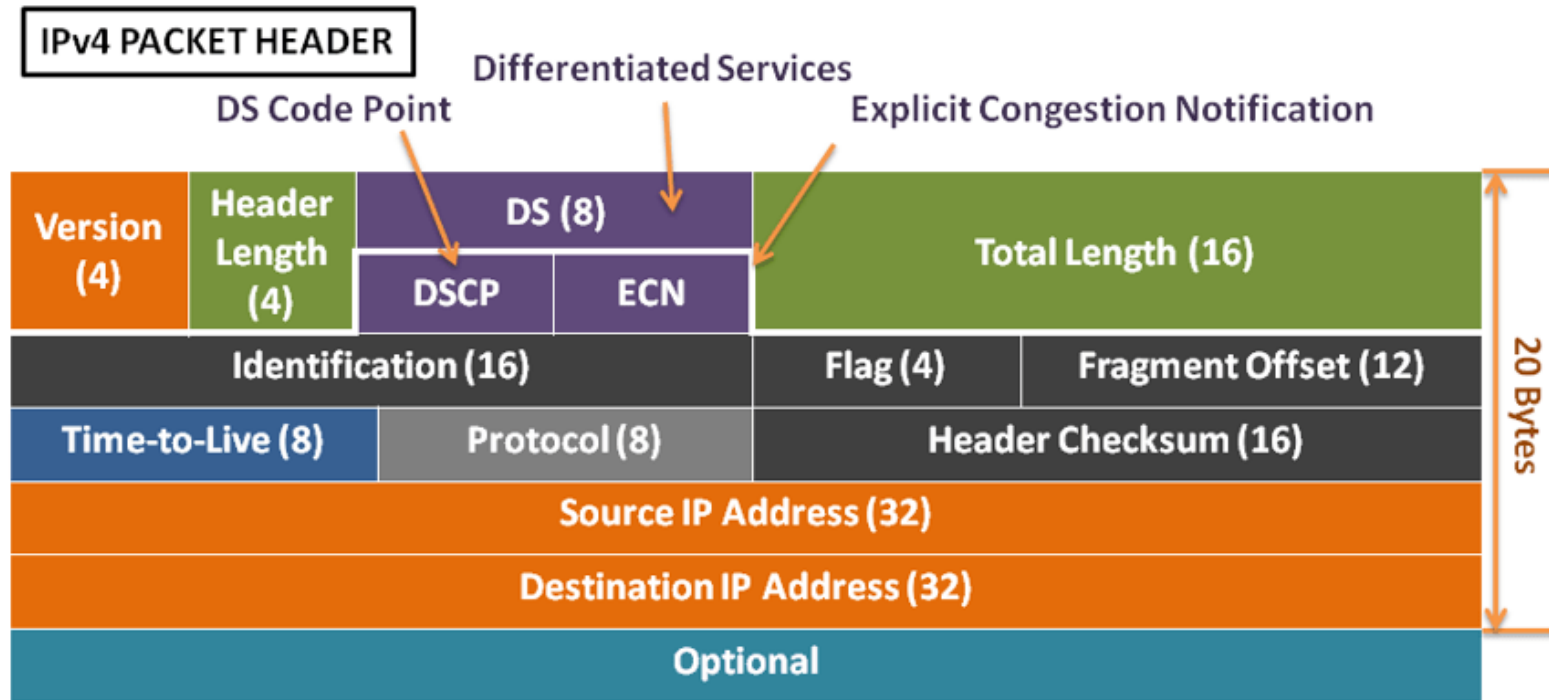
- ❑ Packet switching is cost effective, because switching devices do not need massive amount of secondary storage.
- ❑ Packet switching offers improved delay characteristics, because there are no long messages in the queue(maximum packet size is fixed).
- ❑ Packet can be rerouted if there is any problem, such as, busy or disabled links.
- ❑ The advantage of packet switching is that many network users can share the same channel at the same time.
- ❑ Packet switching can maximize link efficiency by making optimal use of link bandwidth

Disadvantages

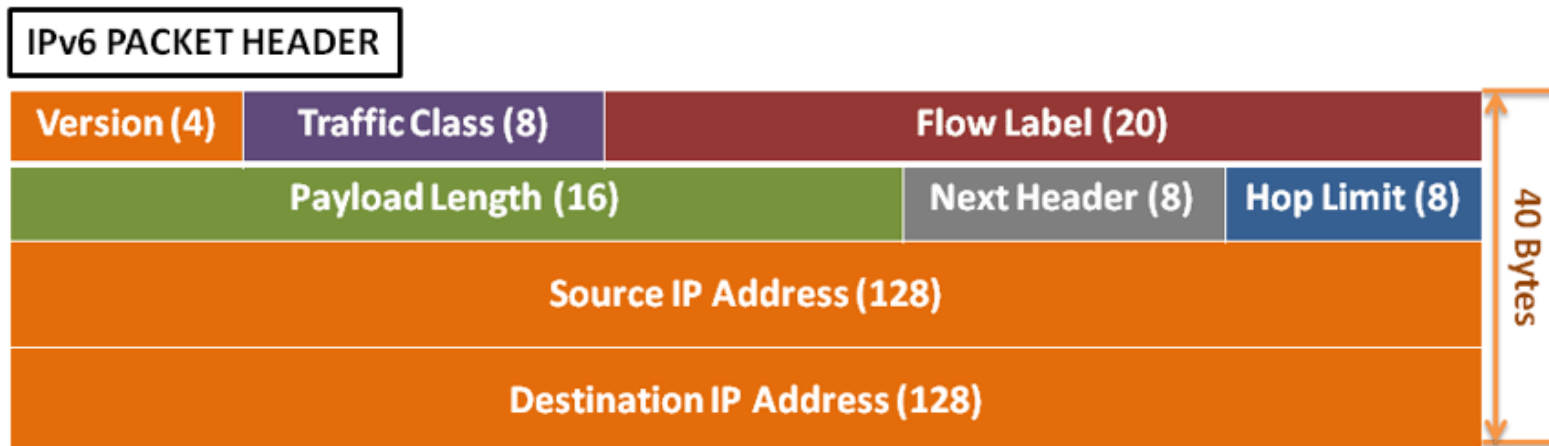
- ❑ Protocols for packet switching are typically more complex.
- ❑ It can add some initial costs in implementation.
- ❑ If packet is lost, sender needs to retransmit the data.
- ❑ Another disadvantage is that packet-switched systems still can't deliver the same quality as dedicated circuits in applications requiring very little delay - like voice conversations or moving images.

IP Protocol

- ❑ Generally, Internet Protocol is the primary protocol of the network layer. It uses a best-effort mechanism to transport packets from source to destination.
- ❑ In a network, IP provides only those functions which are required to send packets from source to destination.
- ❑ IP does not care whether the network is wired or wireless. It cares about the delivery of packets. Also, it does not guarantee that the packets sent by the sender will be received correctly by the receiver.
- ❑ IP does not track and maintain the flow of packets.
- ❑ On the sender side, when the network layer receives the segment from the transport layer, it adds the IP header to the segment and forms the packet. Then, the sender sends the packet to the receiver.
- ❑ When the receiver receives the packet, its network layer forwards the packet to the transport layer, and the transport layer decides to which process the packet should be assigned.
- ❑ Basically, there are two versions of IP, one is IPv4 (IP version-4), and the other is IPv6 (IP version-6).



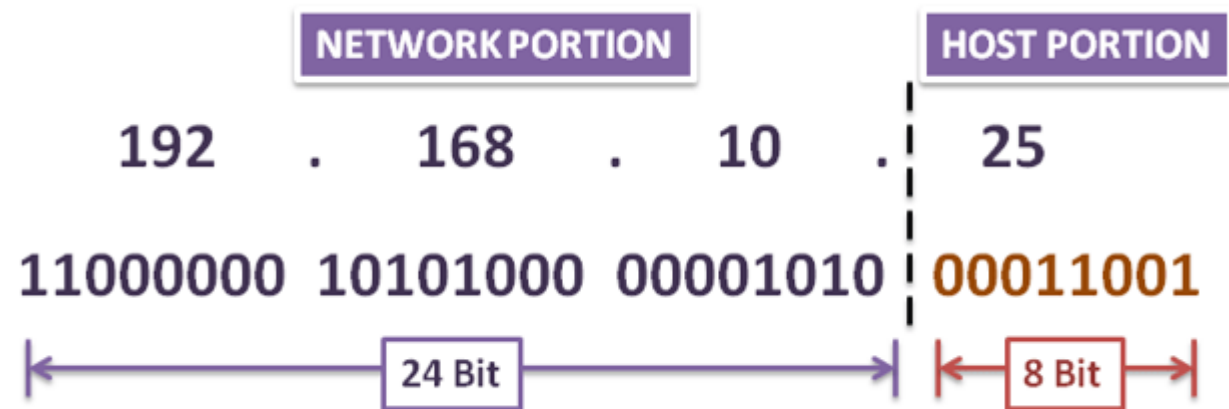
As shown in the figure, Version, Differentiated Services (DS), Header Checksum, Time to Live (TTL), Protocol, and Source and Destination IPv4 Address are the fields of an IPv4 packet.



Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, and Source and Destination IPv6 addresses are the fields of an IPv6 packet.

IPv4 Addressing

- ❑ An IPv4 address is 32-bit number which means that there are a total of 2^{32} (4,294,967,296) IPv4 addresses.
- ❑ Here 4 means there are 4 octets separated by a dot.
- ❑ IPv4 consists of two portions, a network portion and a host portion.
- ❑ The size of the network and host portion varies by network.
- ❑ In our case, 24 bits belong to the network portion, and the remaining 8 bits belong to the host portion.
- ❑ The network portion is the same for all the devices residing in the same network.
- ❑ Each device on the network has a unique IPv4 address.



Subnet Mask

- ❑ When a device receives a unique IPv4 address, it has a subnet mask that separates the network and host portions of the IPv4 address.
- ❑ Using the subnet mask, the router determines the network address of the device.
- ❑ The subnet mask guides the router to know how long the network portion is.
- ❑ The network portion and the host portion can be identified by the AND method.

	11000000 10101000 00001010 00011001	192.168.10.25 – Host IPv4 Address
AND	11111111 11111111 11111111 00000000	255.255.255.0 – Subnet Mask
	11000000 10101000 00001010 00000000	192.168.10.0 – Network IPv4 Address

Figure 3: AND Operation between Host IPv4 Address and Subnet Mask



IP Addresses

❑ Network address:

- The network address represents a unique address in which multiple devices exist along with the host IP address. In our case, 200.12.1.0/24 is the network address.
- A device belongs to the 200.12.1.0 network if its subnet is the same as the network address.

❑ Host Address:

- Host addresses are assigned to each device. This is unique to each device. The 200.12.1.0 network has a total of 256 IPs which range from 200.12.1.0-200.12.1.255.
- The first address 200.12.1.0 is used for the network address. The last address 200.12.1.255 is used as the broadcast address.
- The addresses from 200.12.1.1 to 200.12.1.254 are used as host addresses. Out of 256 IP addresses, 254 addresses are used as host addresses.

❑ Broadcast Address:

- The last address of the network is used as the broadcast address. Generally, it is used to broadcast packets.
- In a broadcast address, the host bits are all 1. In our example, 200.12.1.255/24 is the broadcast address.

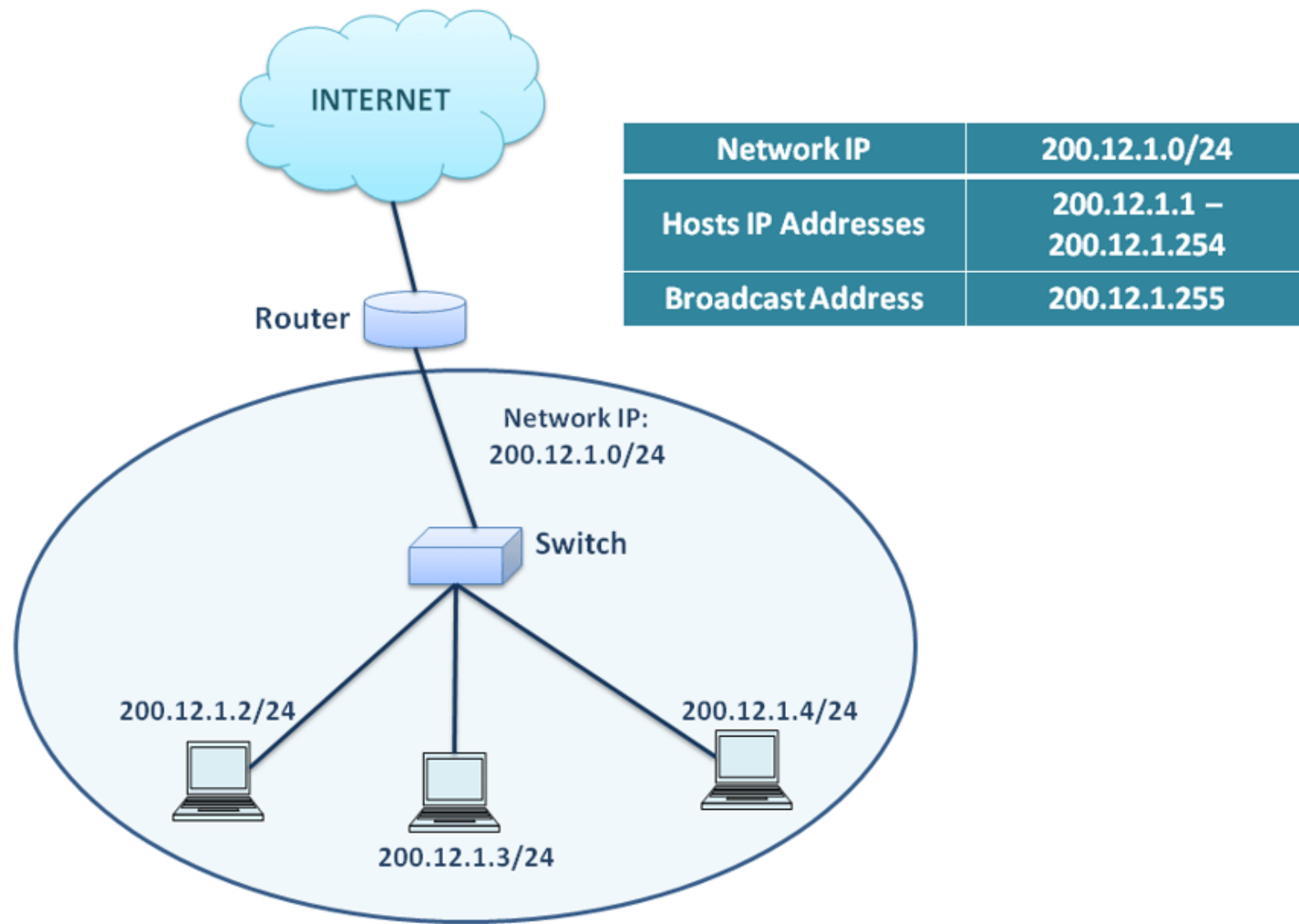


Figure 4: Types of IPv4 Addresses



Calculation of hosts in an IP address

- ❑ The IP address 192.168.12.0 has the /24 subnet mask.
- ❑ The first 24 bits belong to the network portion, and the remaining 8 bits ($32 - 24 = 8$) belong to the host part.
- ❑ So, the total number of hosts for 192.168.12.0 IP address is,
 - Number of hosts = $2^8 - 2$
 - Number of hosts = $256 - 2$
 - Number of hosts = 254
- ❑ Here (-2) indicates that the first address of IP 192.168.12.0/24 defines the 192.168.12.0 network IP and the last IP of 192.168.12.0/24 defines the 192.168.12.255 broadcast address. Both are reserved addresses.

Prefix Length in IPv4 Address

- ❑ We can identify the subnets of IPv4 by using the prefix length. In IP address subnet mask, number of bits set to 1 is called the prefix length.
- ❑ As shown in the figure, the subnet mask is first written in binary format. The number of 1 bit is known as the prefix bit.
- ❑ Basically, the prefix length of the subnet mask is written after the slash notation.
- ❑ Instead of writing like 192.156.12.3 255.255.255.0, the IPv4 address is written as 192.156.12.3/24. Here /24 is 255.255.255.0.

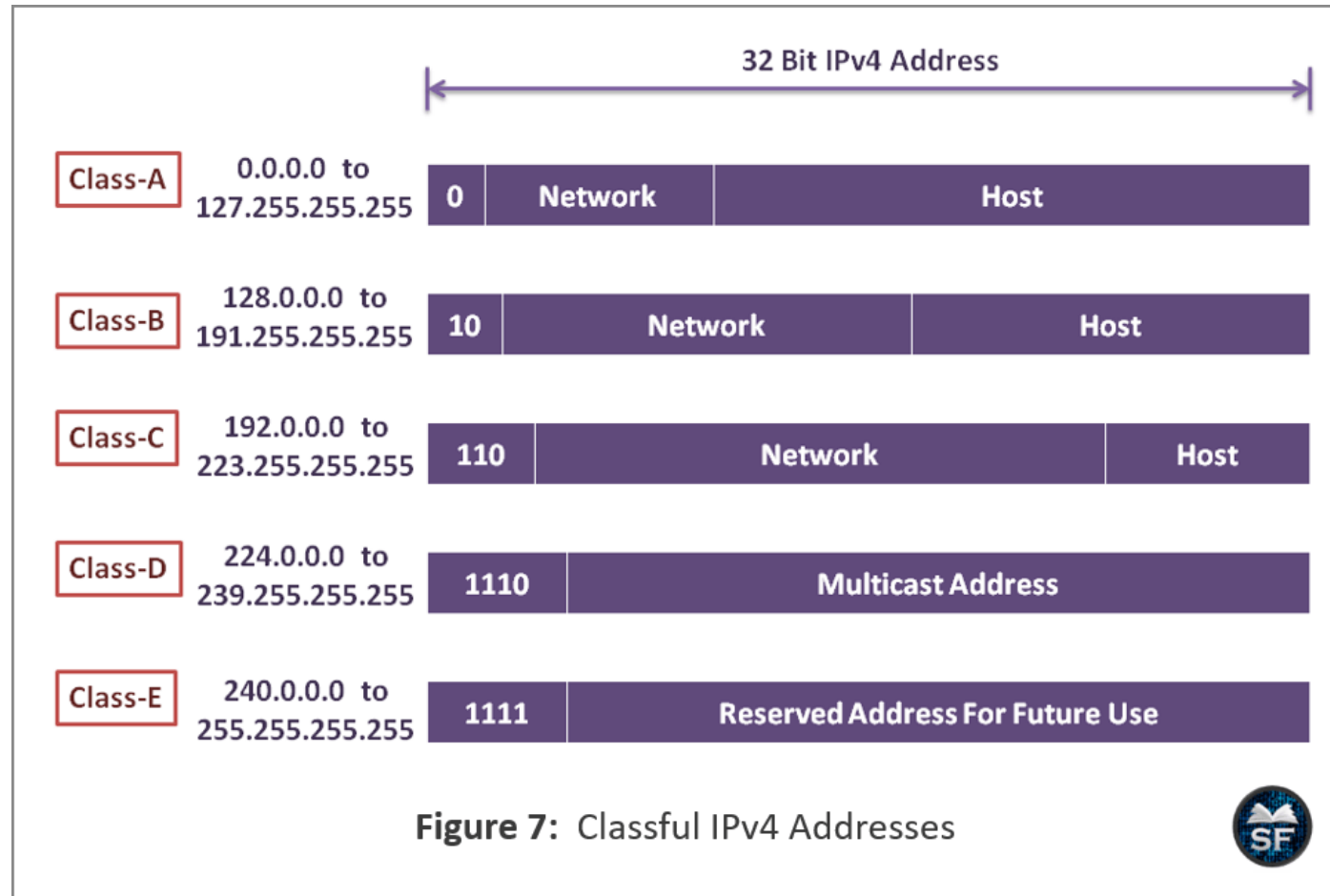
Subnet Mask	IP Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26

Public, Private, and Special Use IPv4 Addresses

- ❑ Public IP Address: Public IP addresses are global, which means that these IP addresses are assigned to any device on the global network.
- ❑ Private IP Address: Private IP addresses are mostly used by organizations. Private IP addresses are allocated to organizations so that communication can take place only in that range of private IP addresses. Basically, the private IPv4 address is not unique. It is used in the internal network.
- ❑ Generally, router didn't communicate using private IP address. It uses network address translation (NAT) to translate the private IP address to the public IP address.
- ❑ Loopback Addresses: Loopback addresses are used when the host wants to send traffic to itself. Addresses ranging from 127.0.0.0/8 or 127.0.0.1 to 127.255.255.254 are known as loopback addresses.
- ❑ Link-Local Addresses: When the device does not get an IP dynamically, it is assigned one of the link-local addresses. It is also known as Automatic Private IP Addressing (APIPA) or Self-Assigned Address.
- ❑ Link-Local Addresses are 169.254.0.0/16 or 169.254.0.1 to 169.254.255.254.

Classful Addressing

- ❑ IPv4 addresses are divided into five classes that are A, B, C, D, and E. Each class has a different number of IPv4 addresses.
- ❑ By looking at the first byte of the IP address, we can determine the class of the IP address.
- ❑ As shown in the figure, class-A IP addresses range from 0 (0.0.0.0/8) to 127 (127.0.0.0/8).
- ❑ Similarly, Class-B, Class-C, Class-D and Class-E IP addresses range from 128 (128.0.0.0/16) to 191 (191.255.0.0/16), 192 (192.0.0.0/24) to 223 (223.255.255.0/24), 224 (224.0.0.0) to 239 (239.0.0.0), and 240 (240.0.0.0) to 255 (255.0.0.0), respectively.
- ❑ It also shows that, as the subnet mask increases, the number of host decreases.



Need For IPv6 Addresses

- ❑ As we know that the population of the world is increasing and the number of devices on the internet is also increasing.
- ❑ Because of this, IPv4 addresses are running out, as they have a total of 4.2 billion IP addresses. So IPv6 is in trend, and it is the successor of IPv4.
- ❑ IPv4 has 32bit address space, whereas IPv6 has 128-bit address space, which means there are 2^{128} IPv6 addresses available.
- ❑ The IETF (Internet Engineering Task Force) developed IPv6 to correct the limitations of IPv4 because there are not enough IPv4 addresses to accommodate the development.
- ❑ There exist some protocols which are used by the network administrator if he wants to migrate his IPv4 address network to the IPv6 address network.
- ❑ The tunneling method is used to transport IPv6 packets over an IPv4 network because the IPv6 packet is encapsulated in an IPv4 packet.
- ❑ It is also possible on the network that a device that has an IPv6 address can communicate with a device with an IPv4 address using the NAT64 (Network Address Translation 64) method.

IPv6 Addressing

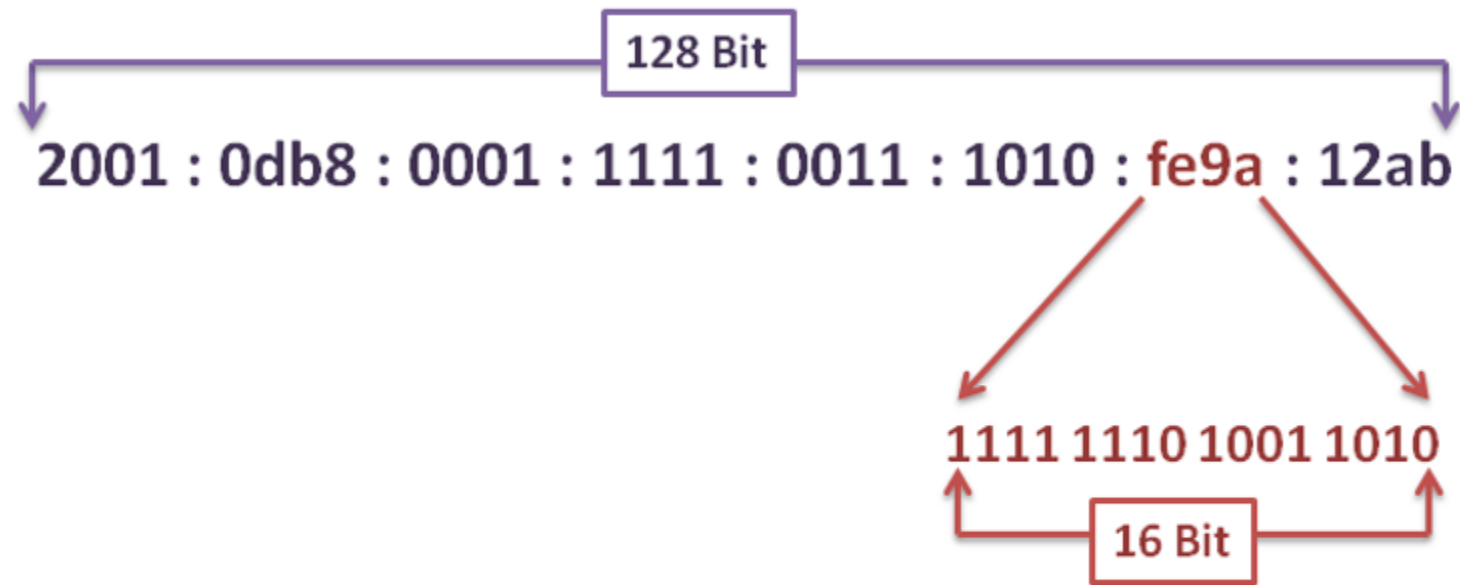


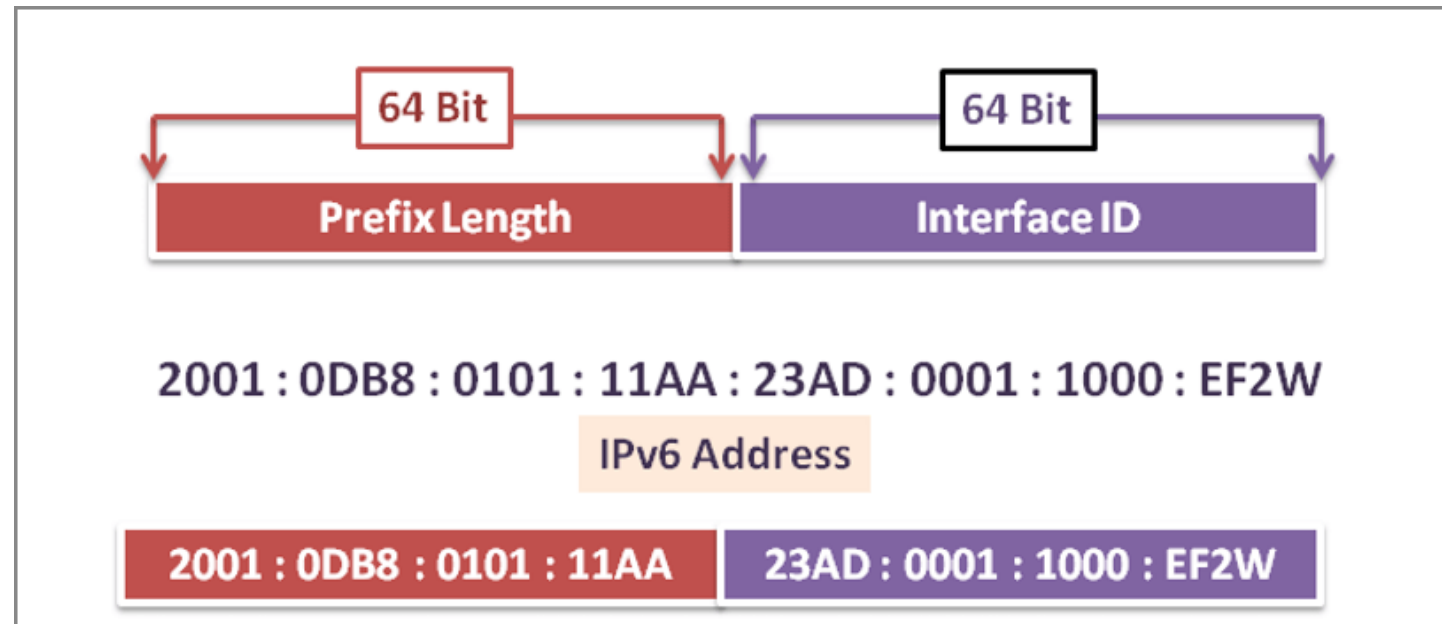
Figure 9: The Structure of an IPv6 Address



-
- ❑ In an IPv6 address, there are 8 sections separated by colons (:), with 16 bits in each section. Let us understand the format of the IPv6 address.
 - ❑ As you can see, there are 8 segments of 16 bit, or you can call it Hextets.
 - ❑ Certain rules are used to shorten IPv6 addresses. They are as follows:
 - Rule 1: Omit the leading zero of the IPv6 address. In this rule, to shorten the notation of IPv6, omit the leading zero in any hextet.
 - Rule 2: In this rule, the double colon (::) replaces any single, continuous string of one or 16-bit hextets containing all zeros.
 - ❑ 2001:0011:0101:11AA:0AB3:0000:0000:123A is an IPv6 address.
 - In the first rule, leading zeros is omitted and the resulting shorten IPv6 address becomes 2001:11:101:11AA:AB3:0:0:123A.
 - Now, as a second rule, replace the continuous string containing zeros with a single colon. 2001:11:101:11AA:AB3:0:0:123A becomes 2001:11:101:11AA:AB3::123A.
 - If there is more than one continuous string of all-0 hextets, use a double colon (::) on the longest string.

IPv6 Prefix Length

- ❑ In IPv4, we identify the network portion by looking at the subnet mask. Similarly, in an IPv6 address, the address is divided into two 64-bit portions.
- ❑ The first 64 bit portion is known as the prefix, and the remaining 64 bit portion is known as the interface ID.

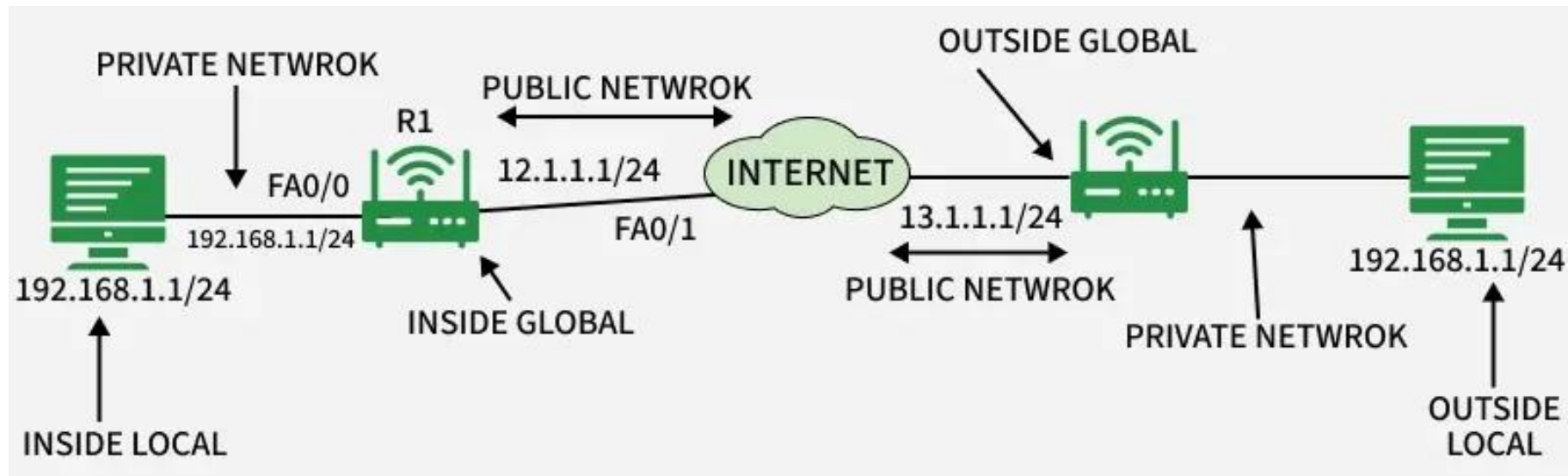
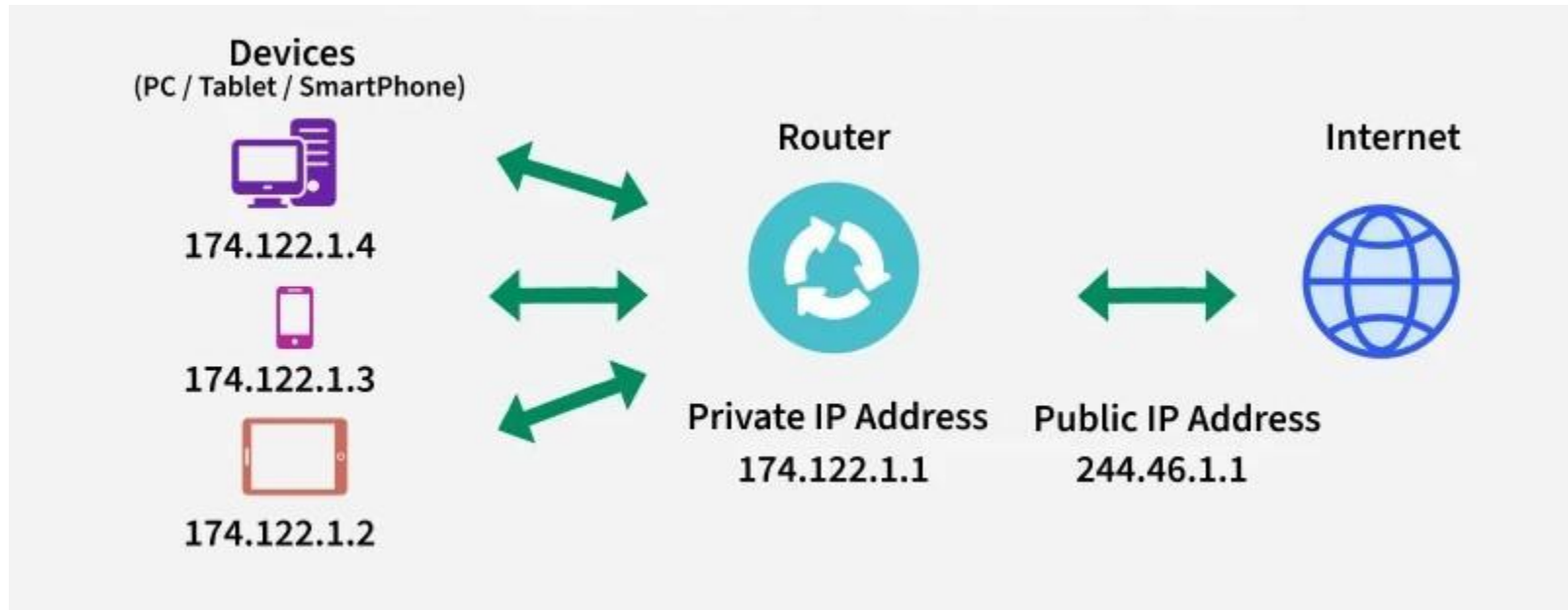


Types of IPv6 Addresses

- ❑ IPv6 unicast addresses are used to identify the interface of a device on a network.
- ❑ IPv6 multicast addresses are used when a packet needs to be sent to multiple devices.
- ❑ IPv6 Unicast Addresses:
 - Global Unicast Address (GUA): GUA is similar to a public IPv4 address in that it is globally unique to the specific device, which can be assigned statically or dynamically. The range of GUA is from 2000::/3 to 3fff::/3.
 - Link-Local Address (LUA): This address is used when two devices are on the same link and is required for each device that has an IPv6 GUA address. Routers do not forward packets using LUA packets. They only send packets using GUA addresses. LLAs are in the range of fe80::/10.
 - Unique Local Address: This address can be used if the device should not be accessible from an external network. Furthermore, it cannot be translated into IPv6 public addresses. They are in the range fc00::/7 to fdff::/7.
 - Loopback Address: The loopback address is the IPv6 address of the device local-host. It is ::1/128. The device uses a loopback address if it wants to send packets to itself.
 - Unspecified Address: As the name suggests IPv6 address is unspecified. Unspecified IPv6 address format is ::.

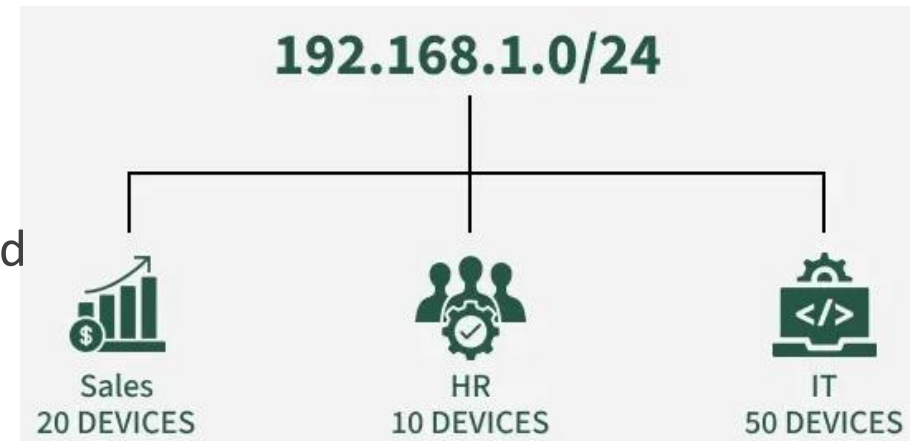
Network Address Translation (NAT)

- ❑ Network Address Translation (NAT) allows multiple devices in a private network to access the internet using a single public IP address.
 - ❑ It helps conserve IPv4 addresses and hides internal systems for added security.
 - ❑ Translates private IPs to public IPs and vice versa.
 - ❑ Prevents IPv4 address exhaustion
 - ❑ Adds security by masking internal devices
 - ❑ Allows thousands of devices to share one public IP
- ❑ Working of NAT
 - This explains how NAT functions:
 - A device sends a request → reaches the NAT-enabled router.
 - Router replaces the private IP with its public IP and assigns a unique port.
 - NAT stores this mapping in the NAT table.
 - When the server responds, NAT uses the stored entry to send the packet to the correct internal device.



Sub-netting

- ❑ Subnetting is the process of dividing a large IP network into smaller logical networks called subnets.
- ❑ Each subnet allows devices to communicate efficiently, improving network performance, security, and manageability.
- ❑ Each department is assigned a separate subnet
- ❑ Devices within a department communicate using their own subnet
- ❑ Inter-department traffic is routed through a router
- ❑ Broadcast traffic is limited to each subnet
- ❑ Network performance, security, and management are improved



Classless Inter Domain Routing (CIDR)

❑ CIDR(Classless Inter Domain Routing) is a method of IP address allocation and routing that allows more efficient use of IP addresses.

❑ Unlike traditional class-based addressing, CIDR allocates IP addresses based on a network prefix rather than a fixed class (A, B, or C).

❑ CIDR Notation: a.b.c.d/n

➤ n = number of bits in the network prefix.

➤ Example: 192.168.1.0/24 first 24 bits are the network, and the remaining 8 bits are the host ID.

❑ Why CIDR?

➤ Classful addressing wastes IP addresses:

Class	IPs Available	Hosts	Example Wastage
A	2^{24}	$2^{24} - 2$	Too large for small orgs
B	2^{16}	$2^{16} - 2$	Wastes 49,150 hosts for 214 needed
C	2^8	$2^8 - 2$	Small networks only

Rules for Forming CIDR Blocks

- ❑ All IPs must be contiguous.
- ❑ Block size must be a power of 2 (2^n) simplifies network division.
- ❑ First IP of block divisible by block size least significant bits of host ID should be 0.
- ❑ Example: If the Block size is 25 then, Host Id will contain 5 bits and Network will contain $32 - 5 = 27$ bits.
- ❑ First IP address of the Block must be evenly divisible by the size of the block. in simple words, the least significant part should always start with zeroes in Host Id. Since all the least significant bits of Host Id is zero, then we can use it as Block Id part.



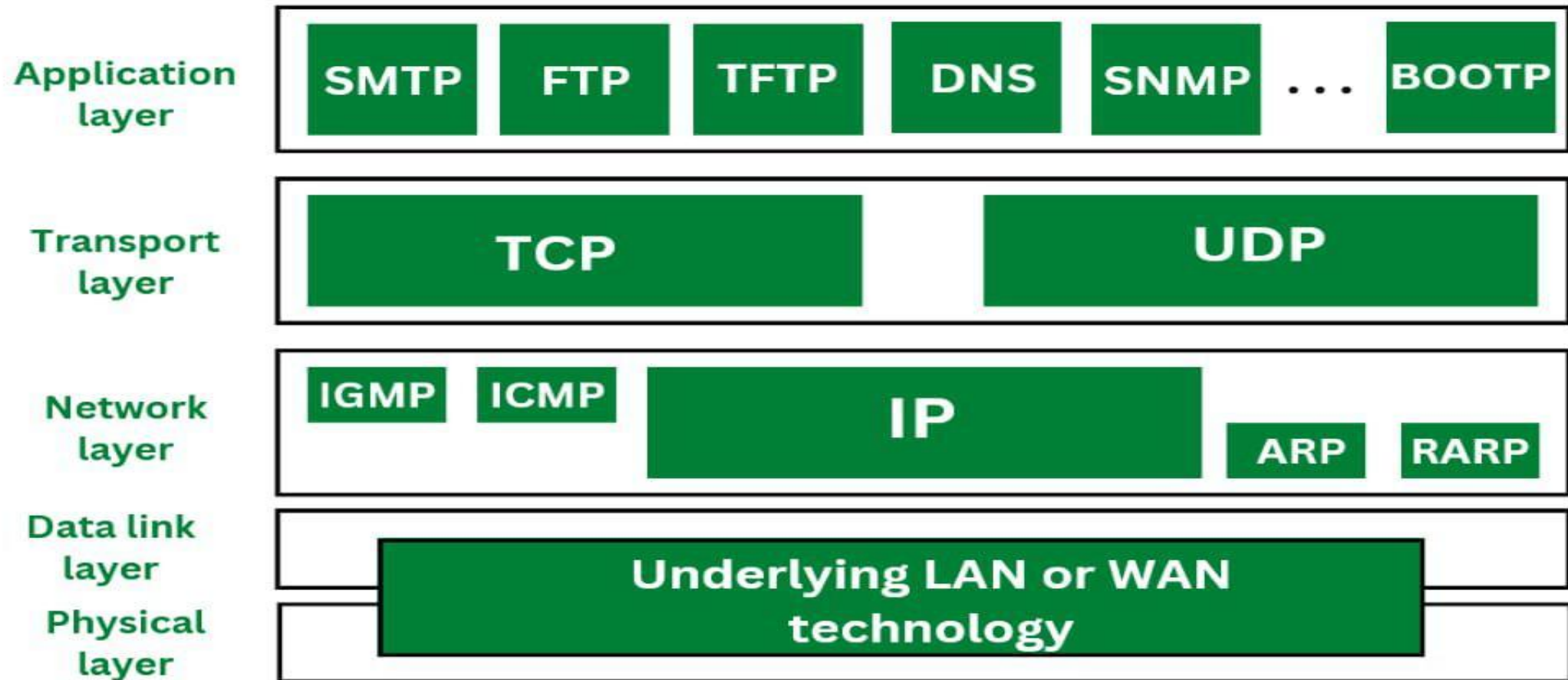
❑ Advantages of CIDR

- Efficient IP usage: Minimizes IPv4 address wastage.
- Flexible allocation: Supports networks of any size.
- Improved routing: Aggregates addresses for faster, simpler routing.
- Lower administrative overhead: Easier IP and network management.

❑ Disadvantages of CIDR

- Complexity: CIDR is more complex to implement and manage compared to traditional class-based addressing.
- Compatibility Issues: Some older network devices may not support CIDR.
- Security Concerns: Implementing security measures like firewall rules and access control lists can be more difficult.

Network layer Protocols

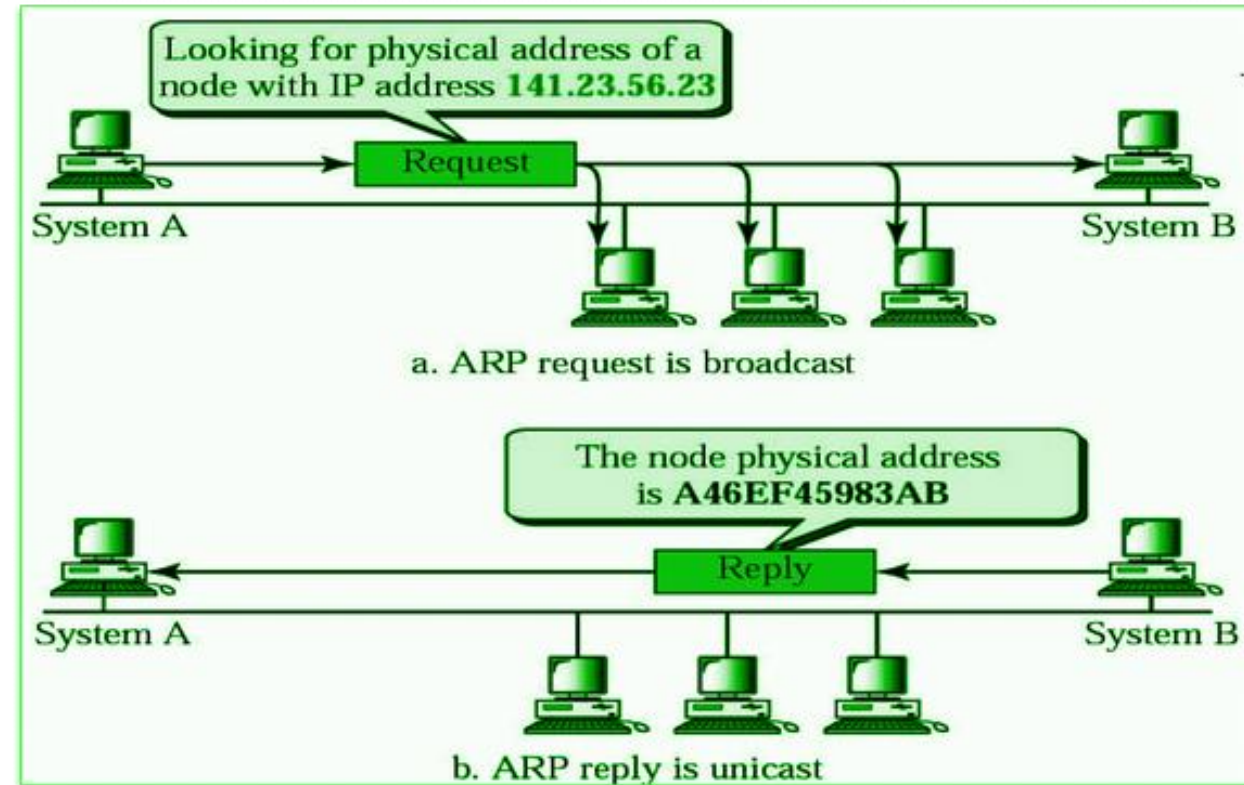


Address Resolution Protocol(ARP)

- ❑ ARP is used to convert the logical address ie. IP address into physical address ie. MAC address.
- ❑ While communicating with other nodes, it is necessary to know the MAC address or physical address of the destination node.
- ❑ If any of the node in a network wants to know the physical address of another node in the same network, the host then sends an ARP query packet.
- ❑ This ARP query packet consists of IP address and MAC address of source host and only the IP address of destination host.
- ❑ This ARP packet is then received to every node present in the network.
- ❑ The node with its own IP address recognizes it and sends it MAC address to the requesting node. But sending and receiving such packets to know the MAC address of destination node it increases the traffic load.
- ❑ Therefore in order to reduce this traffic and improve the performance, the systems that makes use of ARP maintain a cache of recently acquired IP into MAC address bindings.

How Does ARP Work?

- ❑ The host broadcasts an ARP inquiry packet containing the IP address over the network in order to find out the physical address of another computer on its network.
- ❑ The ARP packet is received and processed by all hosts on the network; however, only the intended recipient can identify the IP address and reply with the physical address.
- ❑ After adding the physical address to the datagram header and cache memory, the host storing the datagram transmits it back to the sender.



Reverse Address Resolution Protocol (RARP)

- ❑ RARP works opposite of ARP.
- ❑ Reverse Address Resolution Protocol is used to convert MAC address ie. physical address into IP address ie. logical address.
- ❑ RARP provides with a feature for the systems and applications to get their own IP address from a DNS(Domain Name System) or router.
- ❑ This type of resolution is required for various tasks such as executing reverse DNS lookup. As Reverse Address Resolution Protocol works at low level it requires direct network addresses.
- ❑ The reply from the server mostly carries a small information but the 32 bit internet address is used and it does not exploit the full potential of a network such as ethernet.

How Does RARP Work?

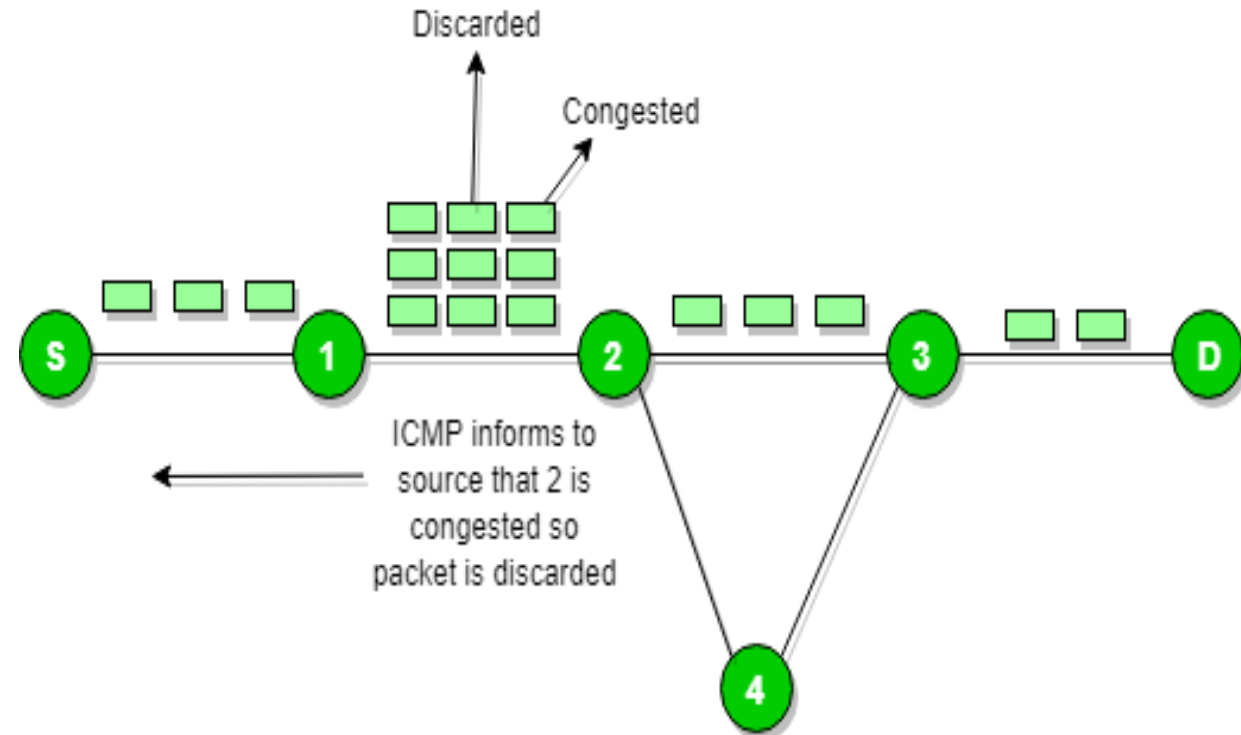
- ❑ Data is sent between two places in a network using the RARP, which is on the Network Access Layer.
- ❑ Every user on the network has two distinct addresses: their MAC (physical) address and their IP (logical) address.
- ❑ Software assigns the IP address, and the hardware then builds the MAC address into the device.
- ❑ Any regular computer connected to the network can function as the RARP server, answering to RARP queries. It must, however, store all of the MAC addresses' associated IP addresses.
- ❑ Only these RARP servers are able to respond to RARP requests that are received by the network. The information package must be transmitted over the network's lowest tiers.
- ❑ Using both its physical address and Ethernet broadcast address, the client transmits a RARP request. In response, the server gives the client its IP address.

Internet Control Message Protocol (ICMP)

- ❑ ICMP is a part of IP protocol suite. ICMP is an error reporting and network diagnostic protocol.
- ❑ Feedback in the network is reported to the designated host. Meanwhile, if any kind of error occur it is then reported to ICMP.
- ❑ ICMP protocol consists of many error reporting and diagnostic messages.
- ❑ ICMP protocol handles various kinds of errors such as time exceeded, redirection, source quench, destination unreachable, parameter problems etc.
- ❑ The messages in ICMP are divided into two types.
 - Error Message: Error message states about the issues or problems that are faced by the host or routers during processing of IP packet.
 - Query Message: Query messages are used by the host in order to get information from a router or another host.

How Does ICMP Work?

- ❑ However, unlike TCP and UDP, ICMP is a connectionless protocol, meaning it doesn't require a connection to be established with the target device in order to transmit a message.
- ❑ TCP and ICMP operate differently from one another; TCP is a connection-oriented protocol, while ICMP operates without a connection.
- ❑ Every time a connection is made prior to a message being sent, a TCP Handshake is required of both devices.
- ❑ Datagrams including an IP header containing ICMP data are used to transmit ICMP packets.
- ❑ An independent data item like a packet is comparable to an ICMP datagram.



Internet Group Message Protocol (IGMP)

- ❑ IGMP is a multicasting communication protocol. It utilizes the resources efficiently while broadcasting the messages and data packets.
- ❑ IGMP is also a protocol used by TCP/IP.
- ❑ Other hosts connected in the network and routers makes use of IGMP for multicasting communication that have IP networks.
- ❑ In many networks multicast routers are used in order to transmit the messages to all the nodes.
- ❑ Multicast routers therefore receives large number of packets that needs to be sent. But to broadcast this packets is difficult as it would increase the overall network load.
- ❑ Therefore, IGMP helps the multicast routers by addressing them while broadcasting.
- ❑ As multicast communication consists of more than one senders and receivers the Internet Group Message Protocol is majorly used in various applications such as streaming media, web conference tools, games, etc.

How Does IGMP Work?

- ❑ Devices that can support dynamic multicasting and multicast groups can use IGMP.
- ❑ The host has the ability to join or exit the multicast group using these devices. It is also possible to add and remove customers from the group using these devices.
- ❑ The host and local multicast router use this communication protocol. Upon creation of a multicast group, the packet's destination IP address is changed to the multicast group address, which falls inside the class D IP address range.

Network Routing Concepts

❑ Network Routing

- Network routing is the process of selecting the best path for data to travel across one or more networks.
- It plays a vital role in the internet by ensuring that data packets reach their correct destination efficiently.
- Routing determines the path IP packets follow from source to destination.
- It is used in packet-switched networks like the Internet.
- Different routing methods are used to optimize speed, reliability, and performance.

❑ Router

- A router is a networking device that controls how data moves between different networks by checking destination IP addresses and choosing the best path.
- Forwards data packets between networks
- Works at Layer 3 (Network Layer) of the OSI model
- Selects the best route for data transfer

Static Routing

- ❑ Static routing is a non-adaptive routing method where routes are configured manually by the network administrator.
- ❑ It provides complete control over routing decisions but is best suited for small networks.
- ❑ Routes are added manually
- ❑ Full control over packet routing
- ❑ Simple but hard to manage in large networks

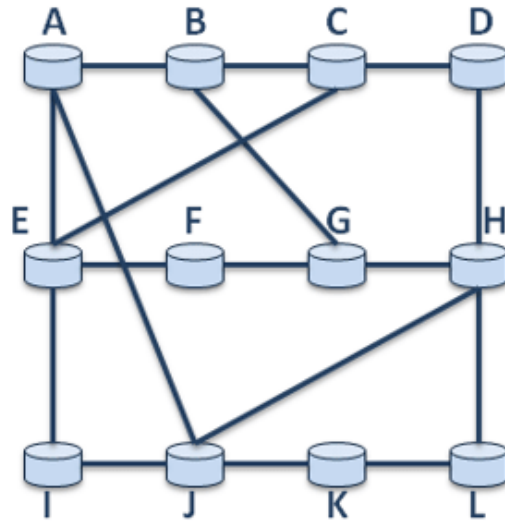
Dynamic Routing

- ❑ Dynamic routing is an automatic and adaptive routing method where routers choose paths using algorithms.
- ❑ It is widely used in modern networks due to its flexibility.
- ❑ Routes update automatically
- ❑ Adapts to network changes
- ❑ Suitable for large networks

Distance Vector Routing (DVR)

- ❑ The distance vector routing algorithm is used to maintain a router's routing table, return the best-known distance to each destination, and find the best route.
- ❑ In a DVR, the routing table is known as a vector. Each router knows the best route to reach each destination.
- ❑ In the DVR algorithm, the routing table has two entries, the outgoing line used for the destination and the estimated distance to the destination.
- ❑ Distance in DVR can be measured as a number of hops or other metric values.
- ❑ If the DVR uses hops as a metric, the distance is one hop. If the metric is the propagation delay, then the distance is the timestamp.

Vectors received from Routers A, I, H, and K to Router-J



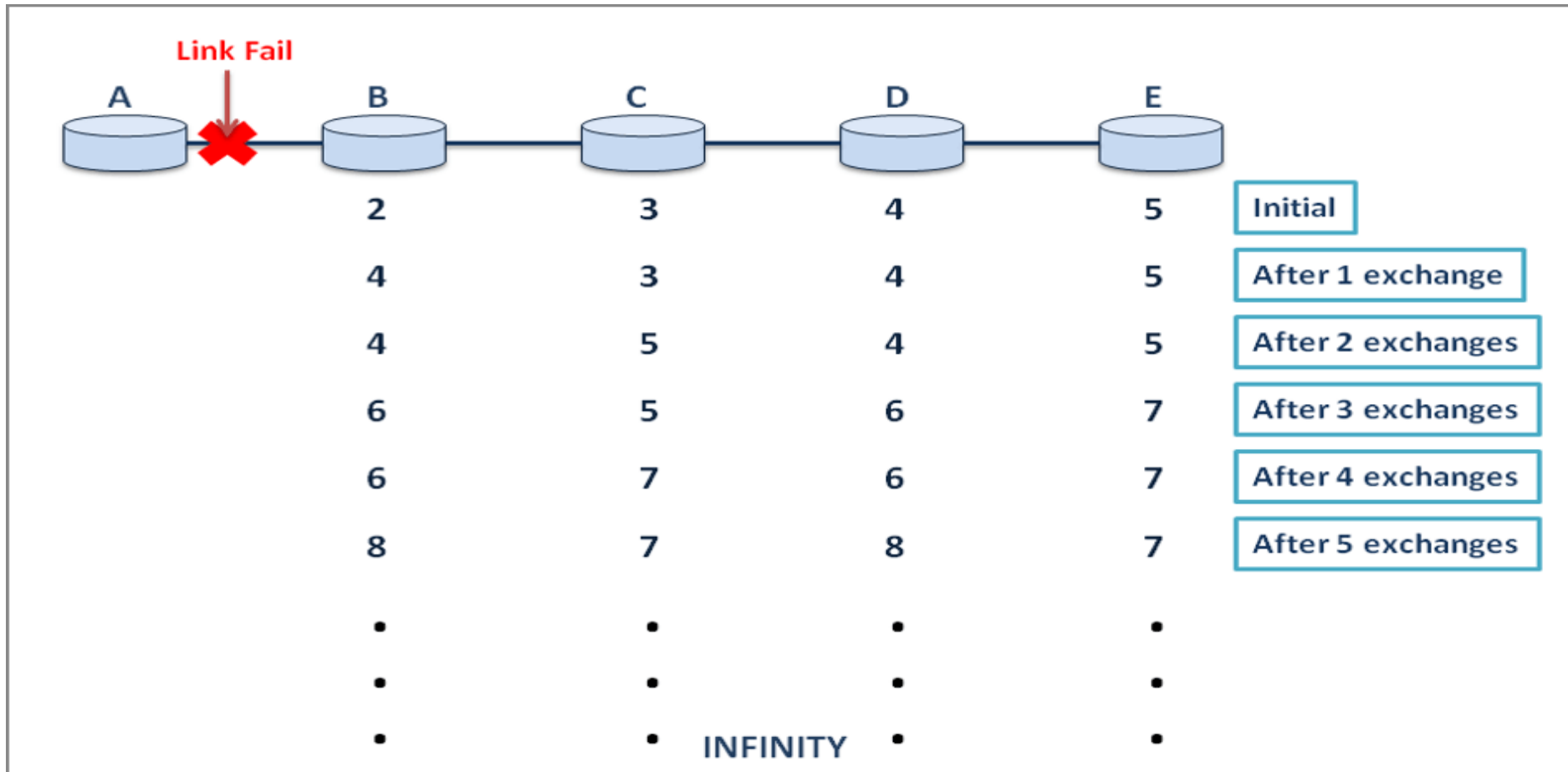
	A	I	H	K	J	
A	0	22	18	19	A	6
B	10	34	29	26	A	18
C	23	16	16	34	I	26
D	38	25	6	22	H	18
E	12	5	28	20	I	15
F	21	18	17	38	I	28
G	16	29	4	29	H	16
H	15	18	0	17	H	10
I	19	0	12	20	I	8
J	7	9	5	8	—	0
K	22	20	20	0	K	4
L	27	31	7	7	K	13
	JA	JI	JH	JK	New routing table for J	

Newly estimated delay from Router-J

Delay: 6 8 10 4

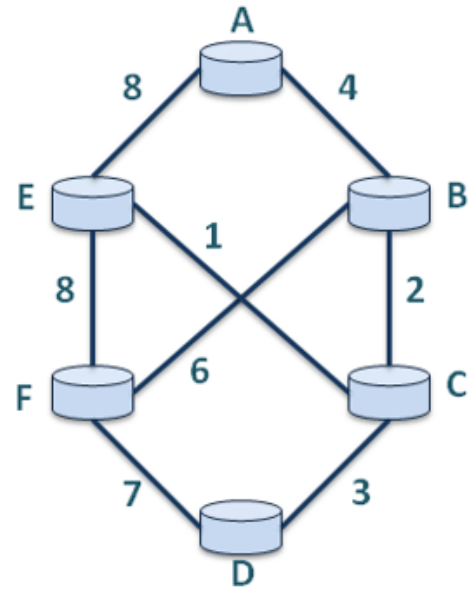
-
- ❑ As shown in the figure, Router-A has 10 msec, 23 msec, 38 msec, etc., respectively, the delay in Router-B, Router-C, Router-D, etc.
 - ❑ Now, Router-J measured the delay in its neighbors Router-A, Router-I, Router-H, and Router-K, which are 6, 8, 10, and 4 msec, respectively.
 - ❑ Router-J wants to communicate with Router-G. Therefore, it has to create a new route for Router-G. Since router-J knows the delays of A, I, H, and K, it tries to find a path through the available router delays.
 - ❑ So, Router-A claims it reaches G in 16 msec, and Router-J reaches Router-A in 6 msec, so Router-J reaches Router-G in $(16+6)$ 22 msec.
 - ❑ Similarly, Router-J through Router-I, Router-H, and Router-K calculates the delays in Router-G as $(29 + 8)$ 37, $(4 + 10)$ 14, and $(29 + 4)$ 33 msec, respectively.
 - ❑ The best and optimal route from Router-J to Router-G is 14, so Router-J makes an entry in its routing table that the delay of G through Router-H is 14 msec.

The Count-to-Infinity Problem



Link State Routing

- ❑ The count-to-infinity problem often takes a long time to converge once the network topology is changed. Consequently, the DVR algorithm is replaced by link-state routing.
- ❑ In link-state routing, each router performs the following steps on the network.
- ❑ Finds neighbors on a network and learns their addresses.
- ❑ Updates the routing table by setting a distance metric for each of its neighbors.
- ❑ Creates a packet and uses all the information received from another router or device to send the packet.
- ❑ Sending packets and receiving packets from other routers.
- ❑ Finds the optimal and shortest path to every other router.



LINK STATE PACKETS

A	
Seq. No.	Age
B	4
E	5

D	
Seq. No.	Age
C	3
F	7

B	
Seq. No.	Age
A	4
C	2
F	6

E	
Seq. No.	Age
A	5
C	1
F	8

C	
Seq. No.	Age
B	2
D	3
E	1

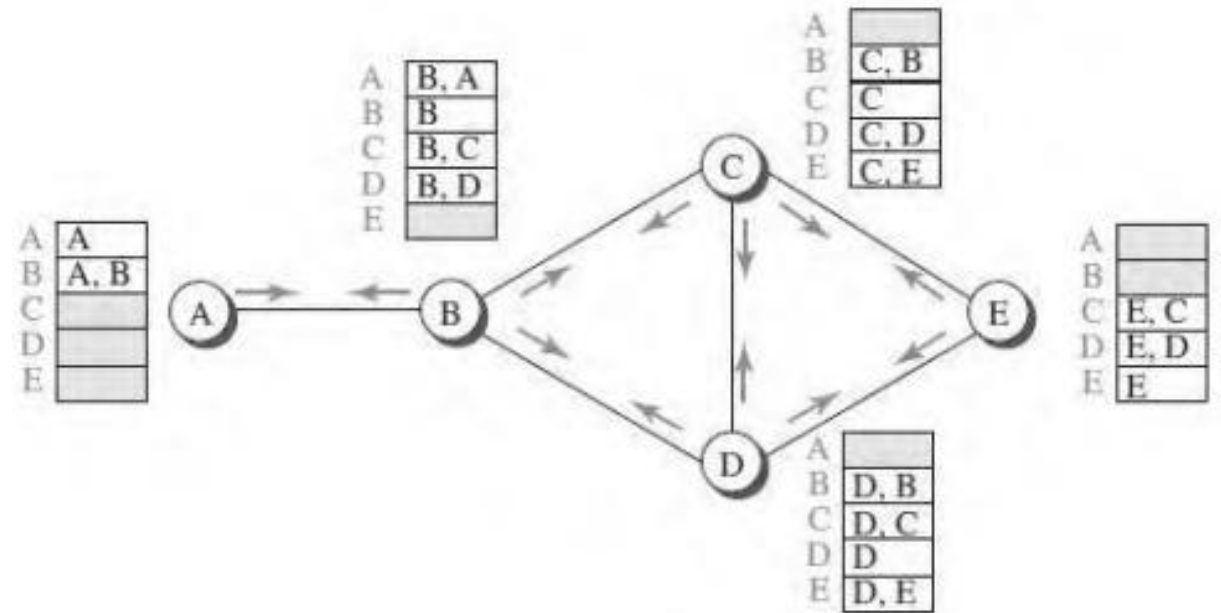
F	
Seq. No.	Age
B	6
D	7
E	8

Figure 4: Link State Packets in LSR



Path Vector

- ❑ Path-vector routing is an asynchronous and distributed routing algorithm.
- ❑ The Path-vector routing is not based on least-cost routing.
- ❑ The best route is determined by the source using the policy it imposes on the route.
- ❑ In other words, the source can control the path.
- ❑ Path-vector routing is not actually used in an internet, and is mostly designed to route a packet between ISPs.



Routing Protocols

RIP

- ❑ The Routing Information Protocol (RIP) is designed to help routers determine the best path for sending data packets across a network. It uses hop count as its routing metric and is primarily used in small to medium-sized networks due to its scalability limitations.
- ❑ RIP operates at the Network Layer (Layer 3) of the OSI model and maintains routing tables on each router.
- ❑ Every 30 seconds, routers exchange their complete routing tables with neighbors using periodic updates.
- ❑ The hop count metric defines the number of routers a packet must pass through to reach its destination.
- ❑ Hop Count
 - Hop count is a routing metric that represents the total number of routers a data packet must pass through to travel from the source device to the destination device.

How RIP Works

- ❑ Routing Information Protocol (RIP) is a distance-vector routing protocol that determines the best path to a destination network based on hop count.
- ❑ Each router maintains a routing table containing distances to all known networks.
- ❑ Routers periodically exchange routing information with neighboring routers.
- ❑ Every 30 seconds, routers broadcast or multicast their entire routing table.
- ❑ If a router learns a shorter path, it updates its routing table accordingly.
- ❑ If a route is not updated within 180 seconds, it is marked as invalid.
- ❑ If the route remains unused for 240 seconds, it is removed (flushed) from the routing table.
- ❑ These timers help maintain accurate routing information and prevent stale routes.

Types of RIP Versions

Feature	RIPv1 (1988)	RIPv2 (1993)	RIPng (1997) for IPv6
Updates sent via	Broadcast (255.255.255.255)	Multicast (224.0.0.9)	Multicast (FF02::9)
Addressing	Classful (no subnet info)	Classless (includes subnet mask)	Classless (IPv6 only)
Authentication	Not supported	Supported	Supported
IP version supported	IPv4 only	IPv4 only	IPv6 only

OSPF

- ❑ Open Shortest Path First (OSPF) is a link-state routing protocol used to find the best path between routers within an Autonomous System (AS).
- ❑ OSPF routers establish neighbor relationships to exchange routing information, and these relationships play a crucial role in the formation of OSPF adjacencies.
- ❑ Understanding the transitions between OSPF protocol states is essential for optimizing network performance and troubleshooting.
- ❑ OSPF uses Hello packets for neighbor discovery and to establish communication between routers.
- ❑ Neighbor states are key in forming OSPF adjacencies, enabling efficient routing.
- ❑ The main focus of this article is to understand OSPF state transitions and their significance in the network.
- ❑ OSPF requires specific parameters like matching Area IDs, authentication, MTU, and timers for adjacencies to form.
- ❑ Distinguishing between OSPF neighbors and adjacent routers is critical for proper network operation.
- ❑ Proper management of OSPF states ensures stable and optimized packet routing.

BGP

- ❑ Border Gateway Protocol (BGP) is an Exterior Gateway Protocol (EGP) used to connect autonomous systems (AS) in any topology.
- ❑ Each AS must have at least one BGP-enabled router connected to another AS's BGP router. Its primary function is to exchange network reachability information between BGP systems.
- ❑ It is a path-vector routing protocol (unlike distance-vector or link-state protocols).

Characteristics of Border Gateway Protocol (BGP)

- ❑ Inter-Autonomous System Configuration: The main role of BGP is to provide communication between two autonomous systems.
- ❑ BGP supports the Next-Hop Paradigm.
- ❑ Coordination among multiple BGP speakers within the AS (Autonomous System).
- ❑ Path Information: BGP advertisements also include path information, along with the reachable destination and next destination pair.
- ❑ Policy Support: BGP can implement policies that can be configured by the administrator.
- ❑ Runs Over TCP.
- ❑ BGP conserves network Bandwidth.
- ❑ BGP supports CIDR.
- ❑ BGP also supports Security.

Open Shortest Path First (OSPF) Terms

- ❑ Router ID: The Router ID is the highest active IP address on the router.
- ❑ Router Priority: The router priority is an 8-bit value assigned to a router running OSPF, which is used during the election of the DR and BDR in a broadcast network.
- ❑ Designated Router (DR): The DR is elected to minimize the number of OSPF adjacencies in a network.
- ❑ Backup Designated Router (BDR): The BDR serves as a backup to the DR in a broadcast network.
- ❑ DR and BDR Election:
 - The election of the DR and BDR occurs in broadcast or multi-access networks. The following criteria are used for the election:
 - The router with the highest priority is elected as the DR.
 - If there is a tie in the router priority, the router with the highest Router ID is chosen. The Router ID is first determined by the highest loopback address, and if no loopback is configured, the highest active IP address on the router's interfaces is considered.

IGRP

- ❑ The Interior Gateway Routing Protocol (IGRP) is a proprietary distance vector routing protocol developed by Cisco Systems in mid-1980s. It uses a distance-vector algorithm to find the best path to a destination in a host network or an autonomous system.
- ❑ Each router in the network maintains a routing table containing information about the best path to reach different networks.
- ❑ Unlike older protocols like Routing Information Protocol (RIP), IGRP overcomes several limitations, particularly the ability to support larger networks and utilize multiple metrics to choose the optimal path.

Features of IGRP Routing Protocol

❑ Composite Metric:

- IGRP uses a composite metric that considers bandwidth, delay (by default), reliability, load, and Maximum Transmission Unit (MTU) to calculate the best path.

❑ Distance Measurement:

- Each hop represents one router between the source and destination. IGRP allows a maximum of 255 hops, far surpassing RIP's limit of 15 hops.

❑ Routing Table:

- Every router maintains a routing table that holds information about the distance to each network and the direction (next hop) to reach that network.

❑ Classful Protocol:

- IGRP is classful, meaning it does not support Variable Length Subnet Masks (VLSM).

❑ Routing Domain:

- All routers within an IGRP network must be in the same Autonomous System (AS).

Quality of Service

- ❑ The quality of service provides reliability to the users by managing the congestion on the network. It focuses on reliability, delay, jitter, and bandwidth.
- ❑ Reliability: there is less reliability or no reliability, the chances of packet or acknowledgment loss increases.
- ❑ Delay: packet delay from the sender to the receiver should be minimal. May tolerate delay in some applications. In telephone communication, audio, or video conferencing, minimal delay is required, whereas in file transfer or email the delay is less significant.
- ❑ Jitter: 's kind of a delay. In audio and video applications, jitter is acceptable. If the difference in latency of the sent packets is high, the jitter is also high.
- ❑ Bandwidth: is used according to the type of network and application. For example, video calling requires more bandwidth because we need to send millions of bits per second to refresh the color screen.